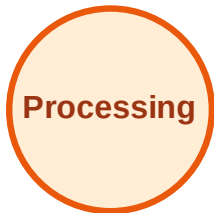


BFS Traversal

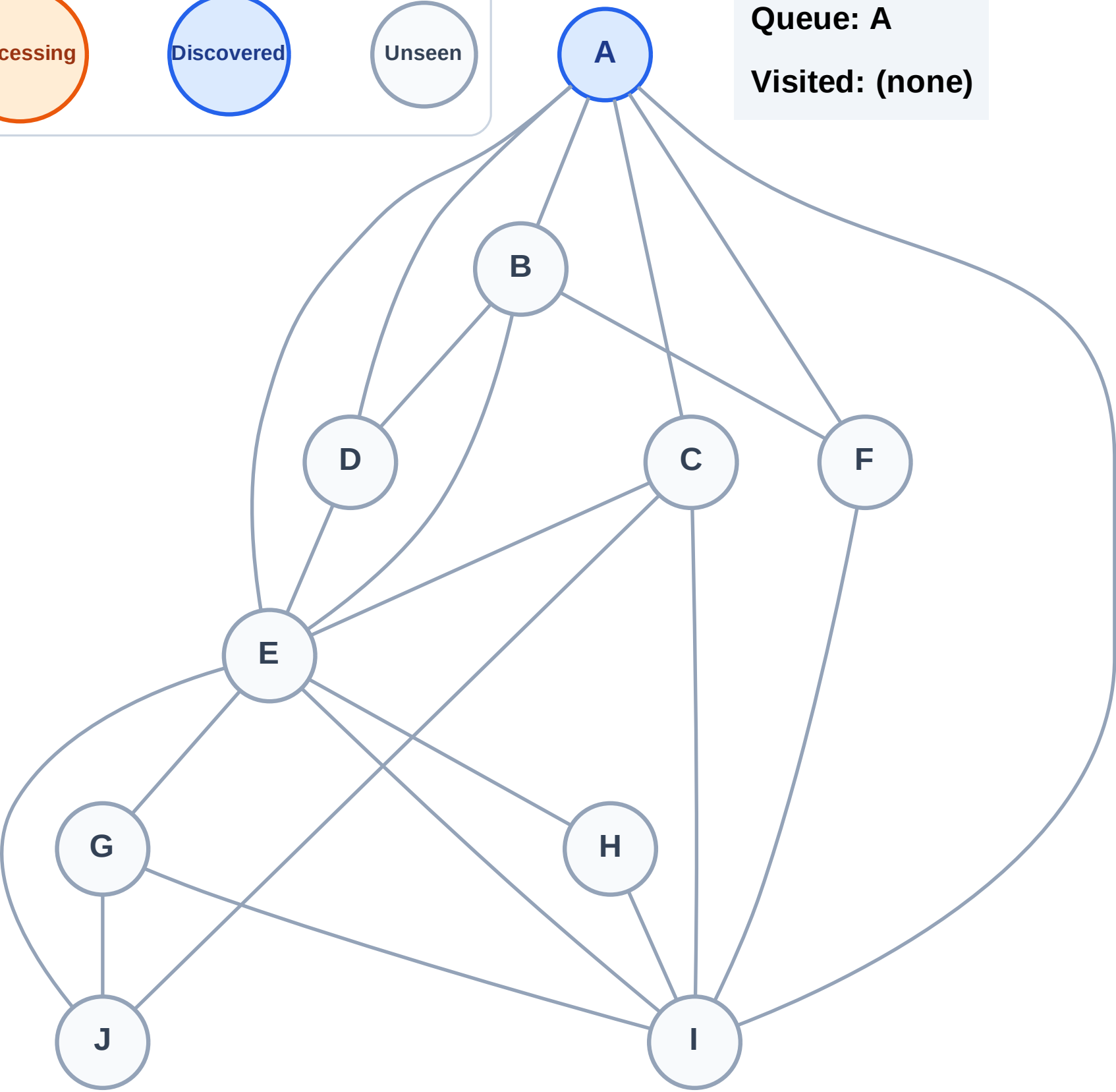
Step 1: Start at A

Legend



Queue: A

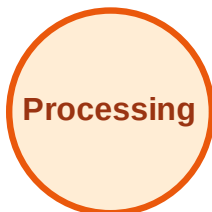
Visited: (none)



BFS Traversal

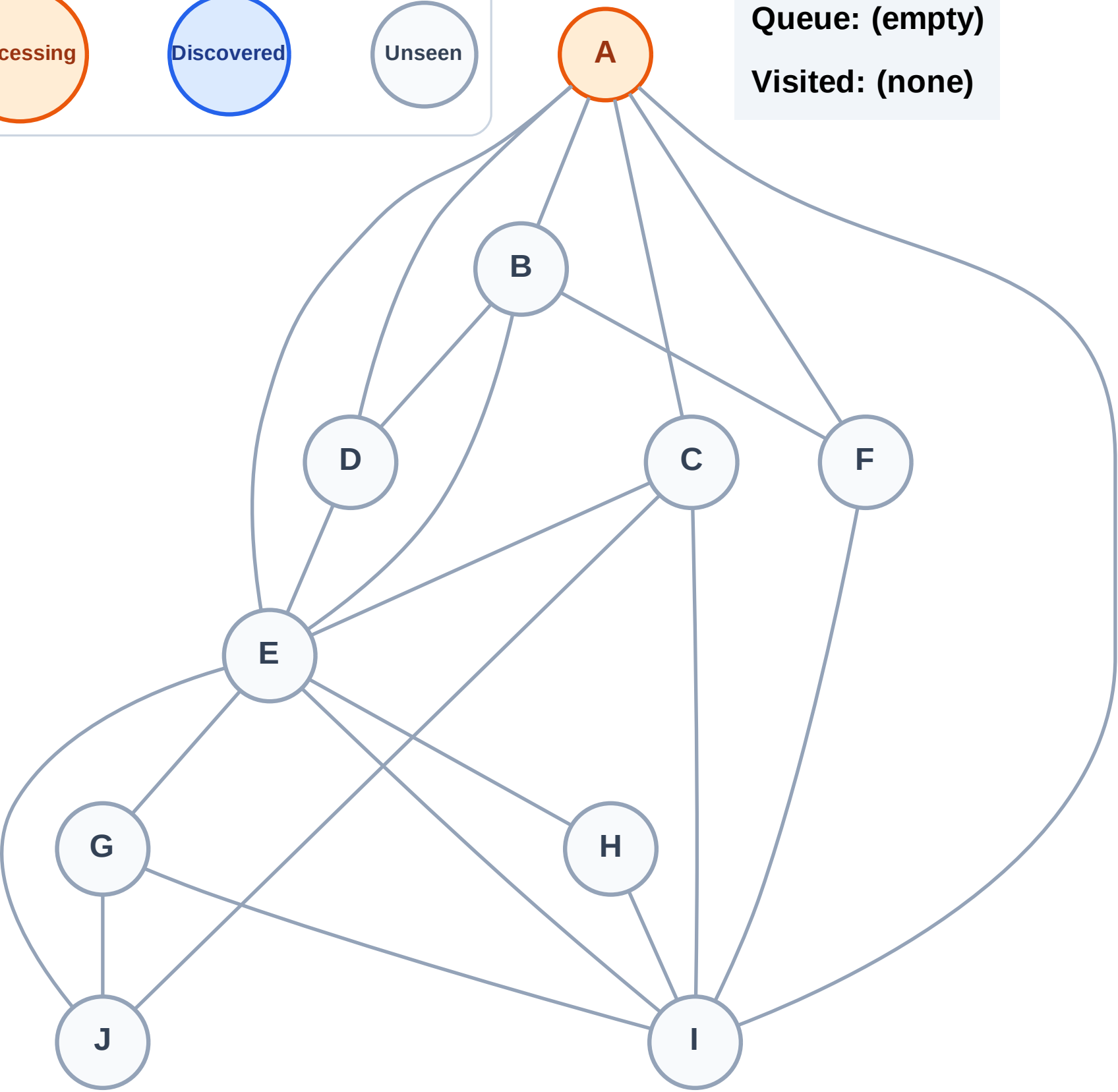
Step 2: Process node A

Legend



Queue: (empty)

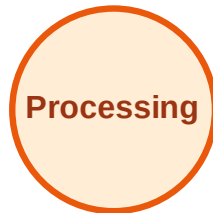
Visited: (none)



BFS Traversal

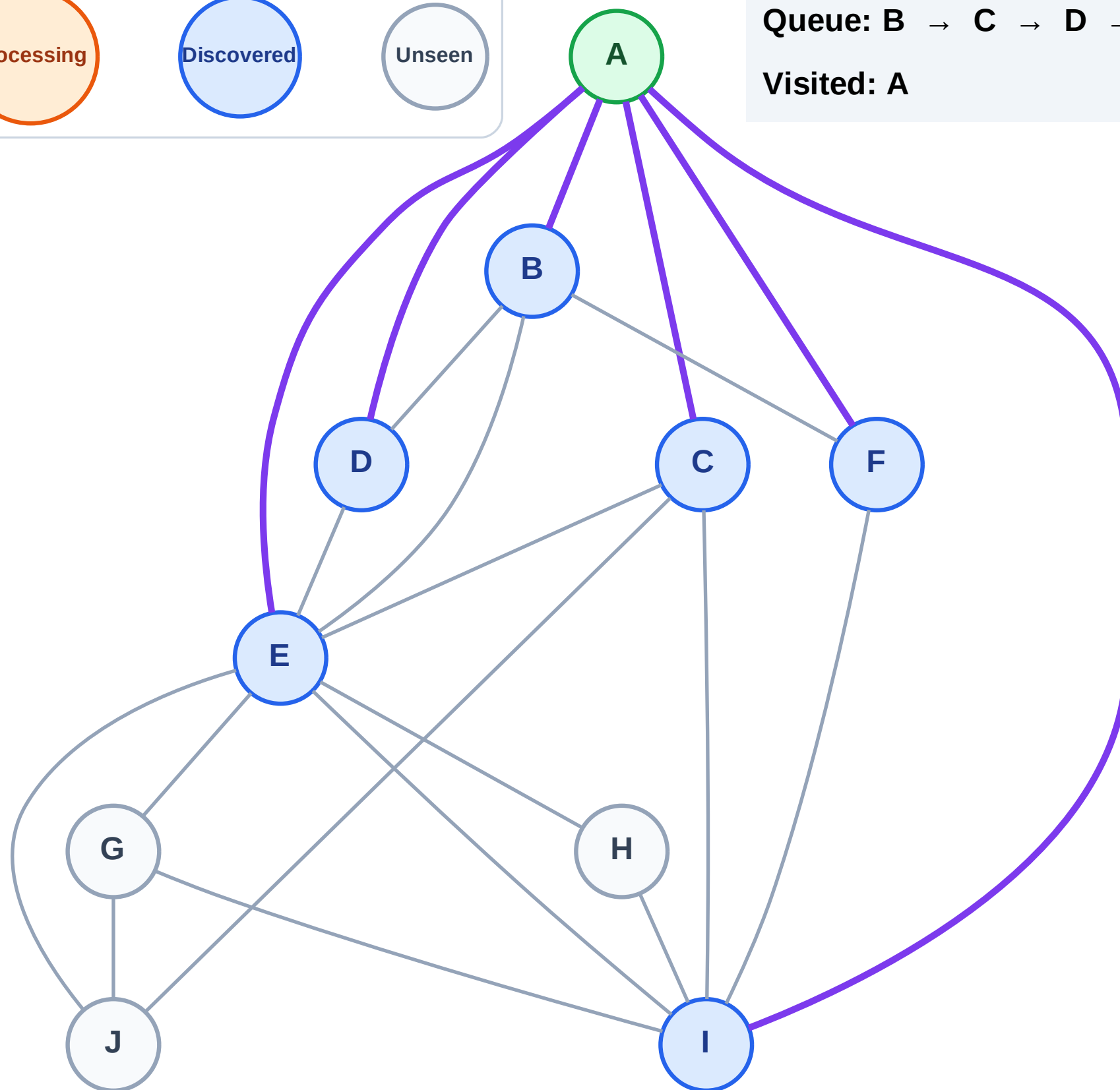
Step 3: Discover B, C, D, E, F, I from A

Legend



Queue: B → C → D → E → F → I

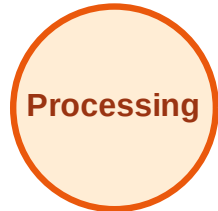
Visited: A



BFS Traversal

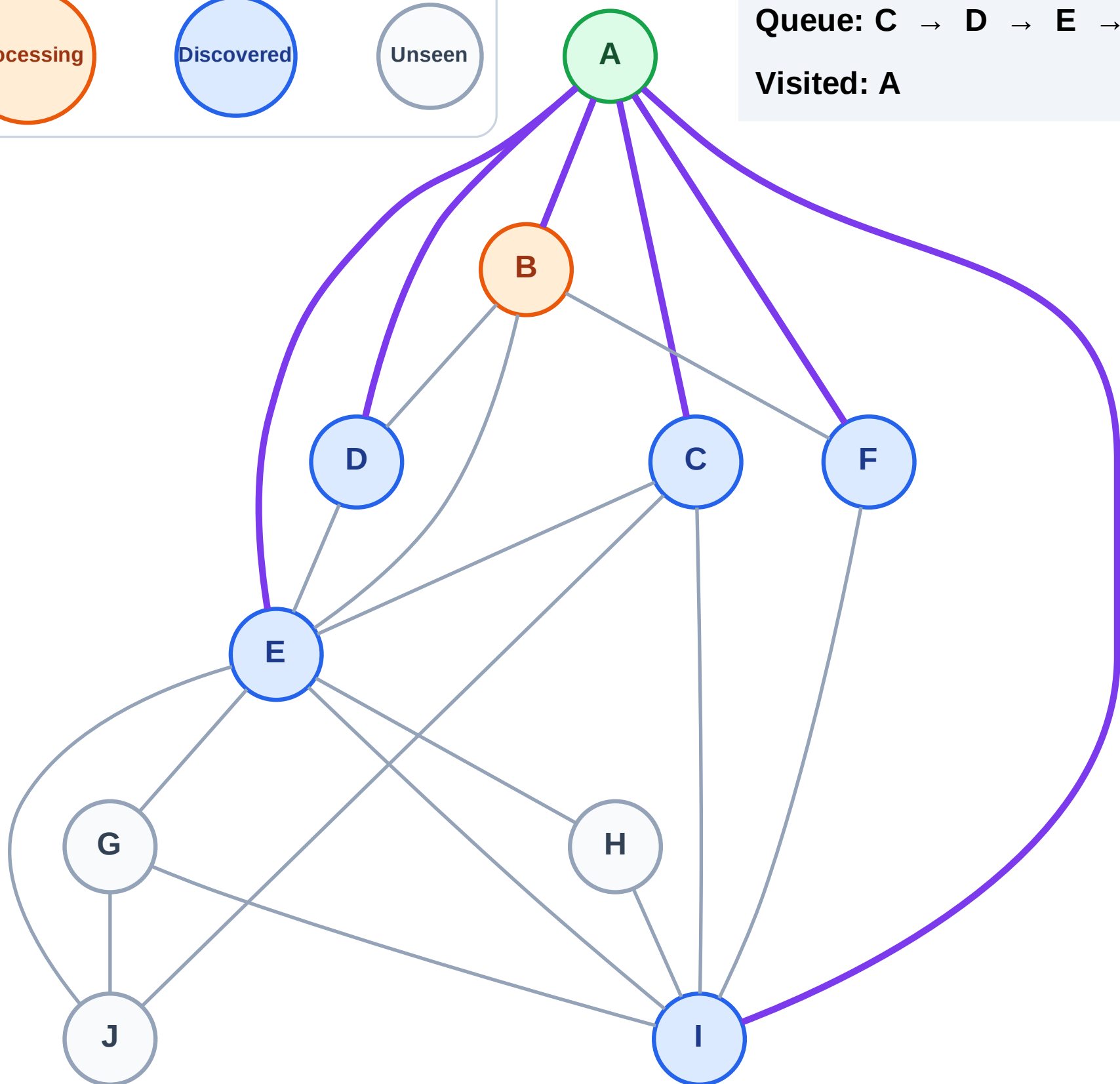
Step 4: Process node B

Legend



Queue: C → D → E → F → I

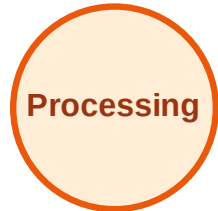
Visited: A



BFS Traversal

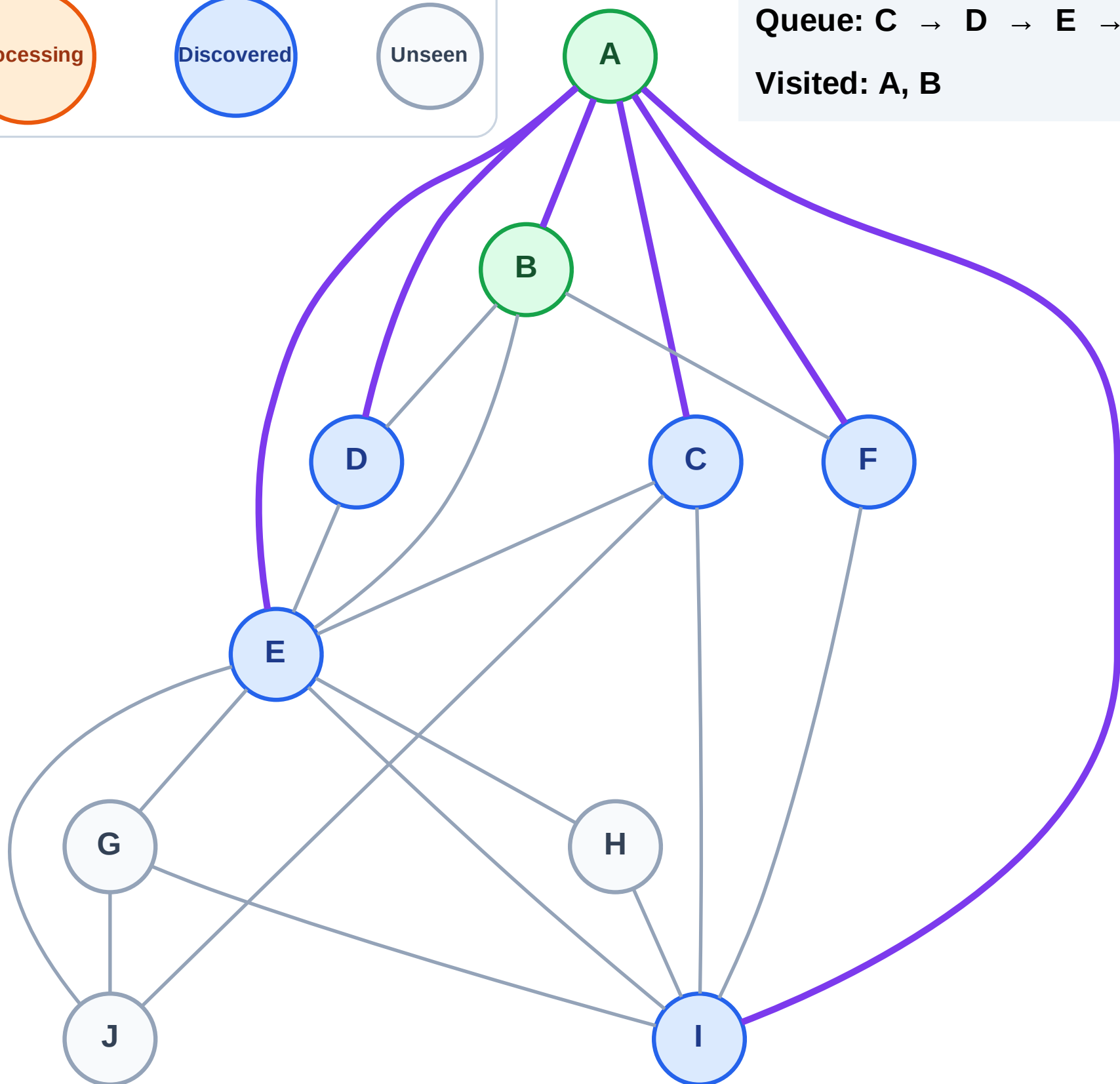
Step 5: No new neighbors from B

Legend



Queue: C → D → E → F → I

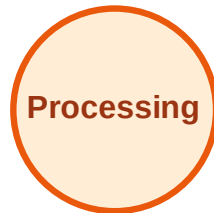
Visited: A, B



BFS Traversal

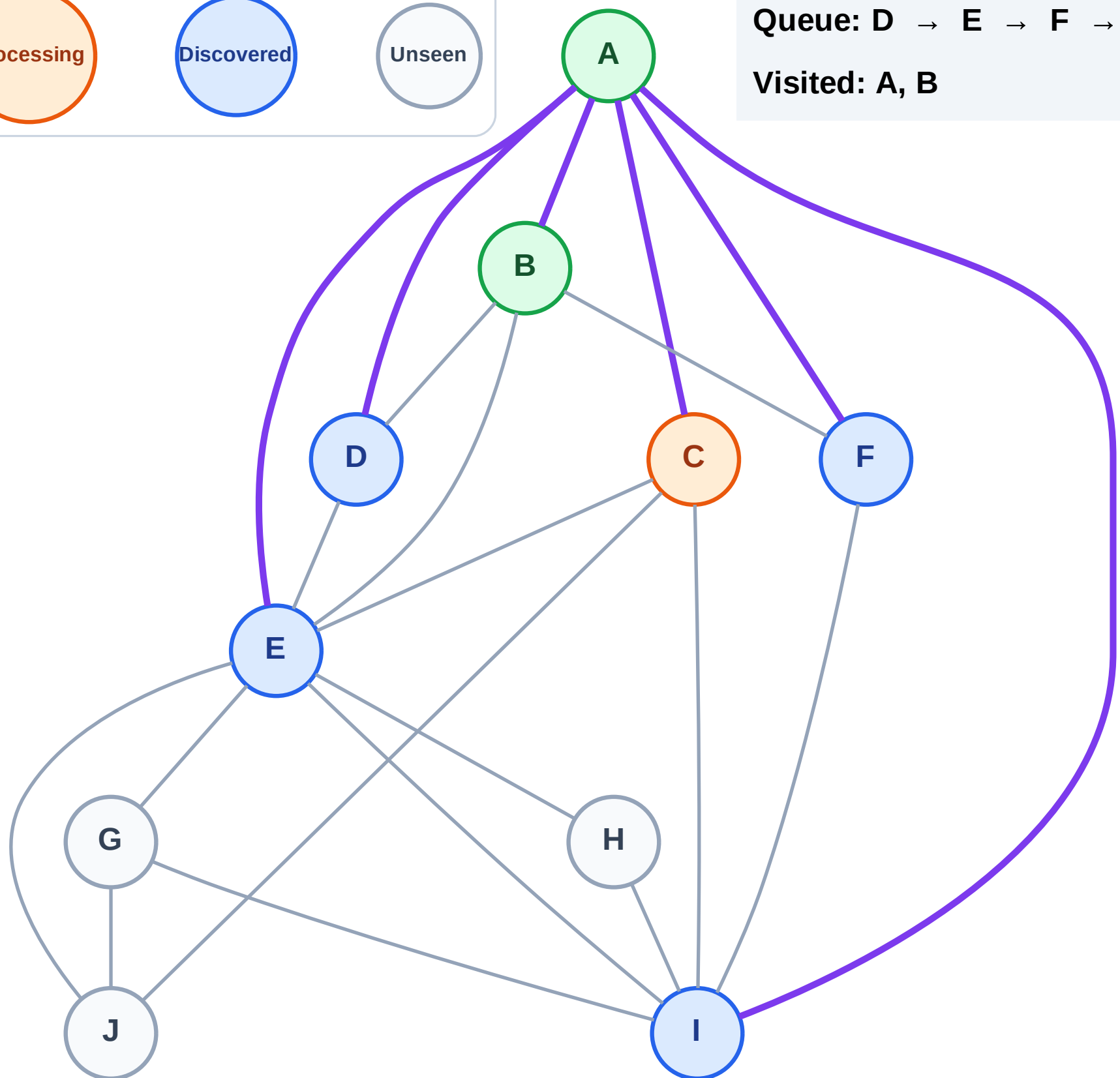
Step 6: Process node C

Legend



Queue: D → E → F → I

Visited: A, B



BFS Traversal

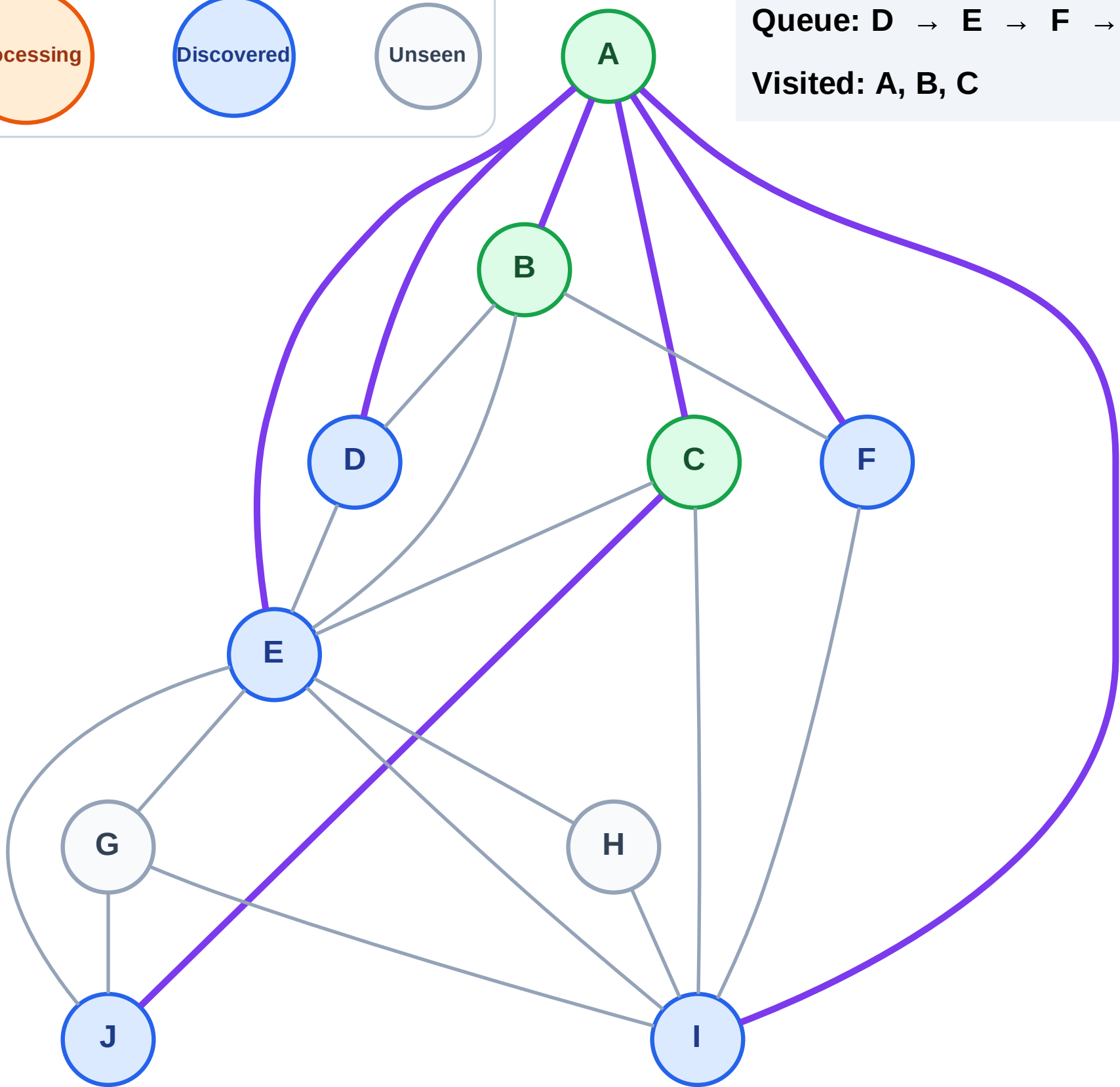
Step 7: Discover J from C

Legend



Queue: D → E → F → I → J

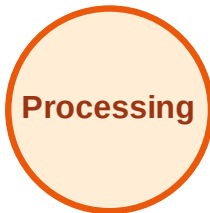
Visited: A, B, C



BFS Traversal

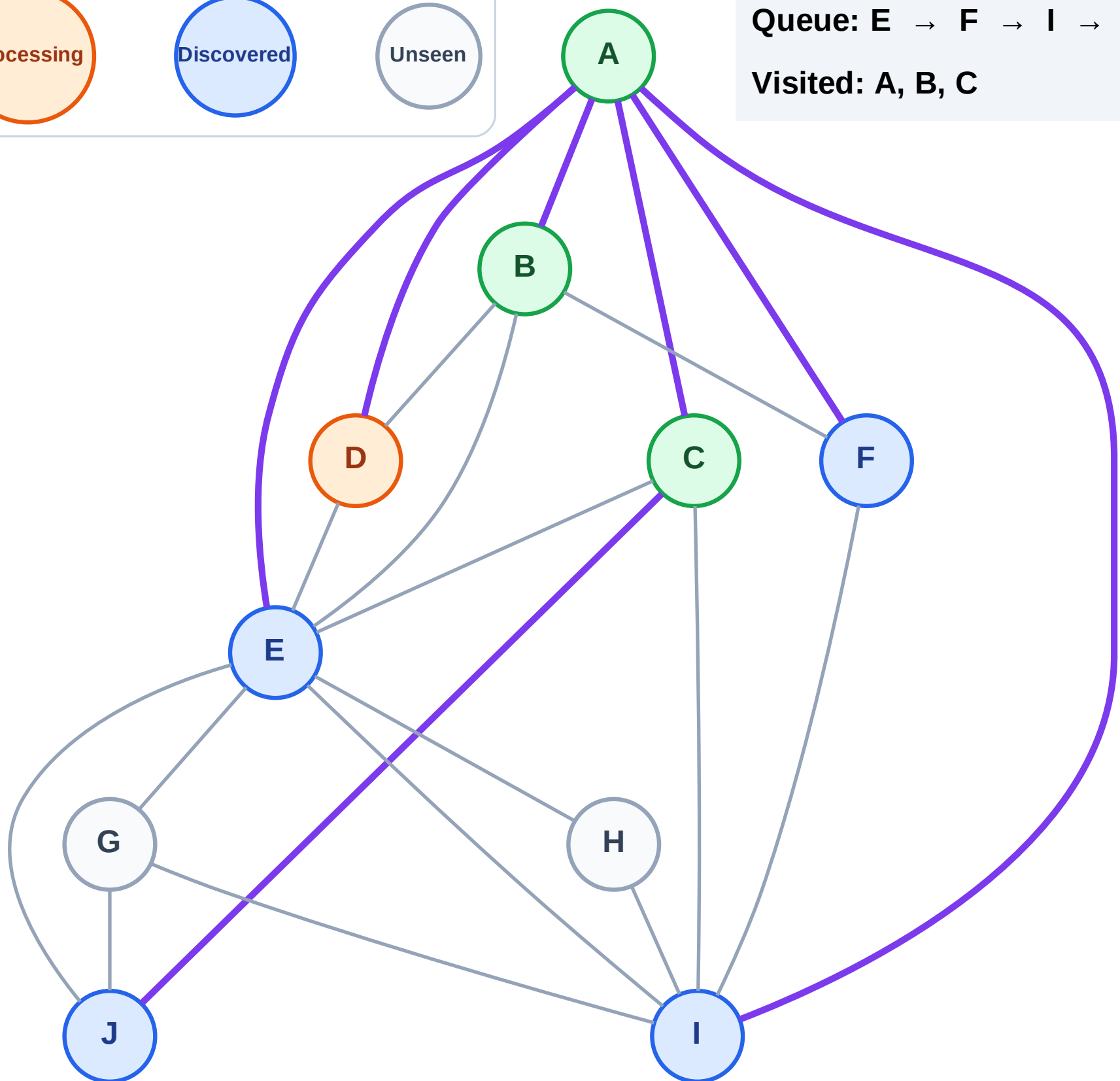
Step 8: Process node D

Legend



Queue: E → F → I → J

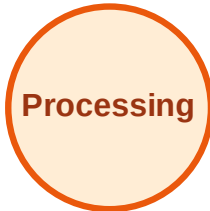
Visited: A, B, C



BFS Traversal

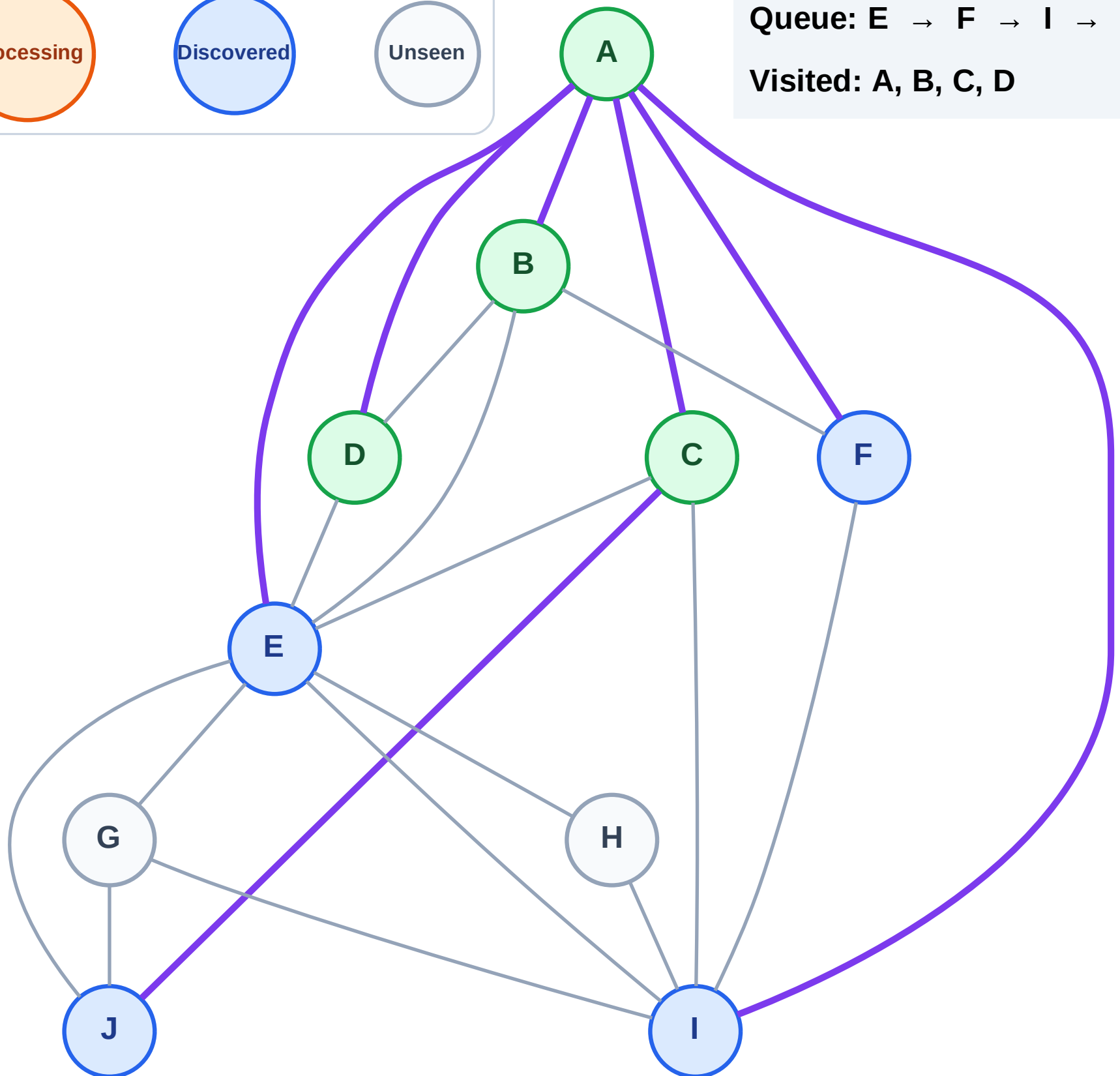
Step 9: No new neighbors from D

Legend



Queue: E → F → I → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

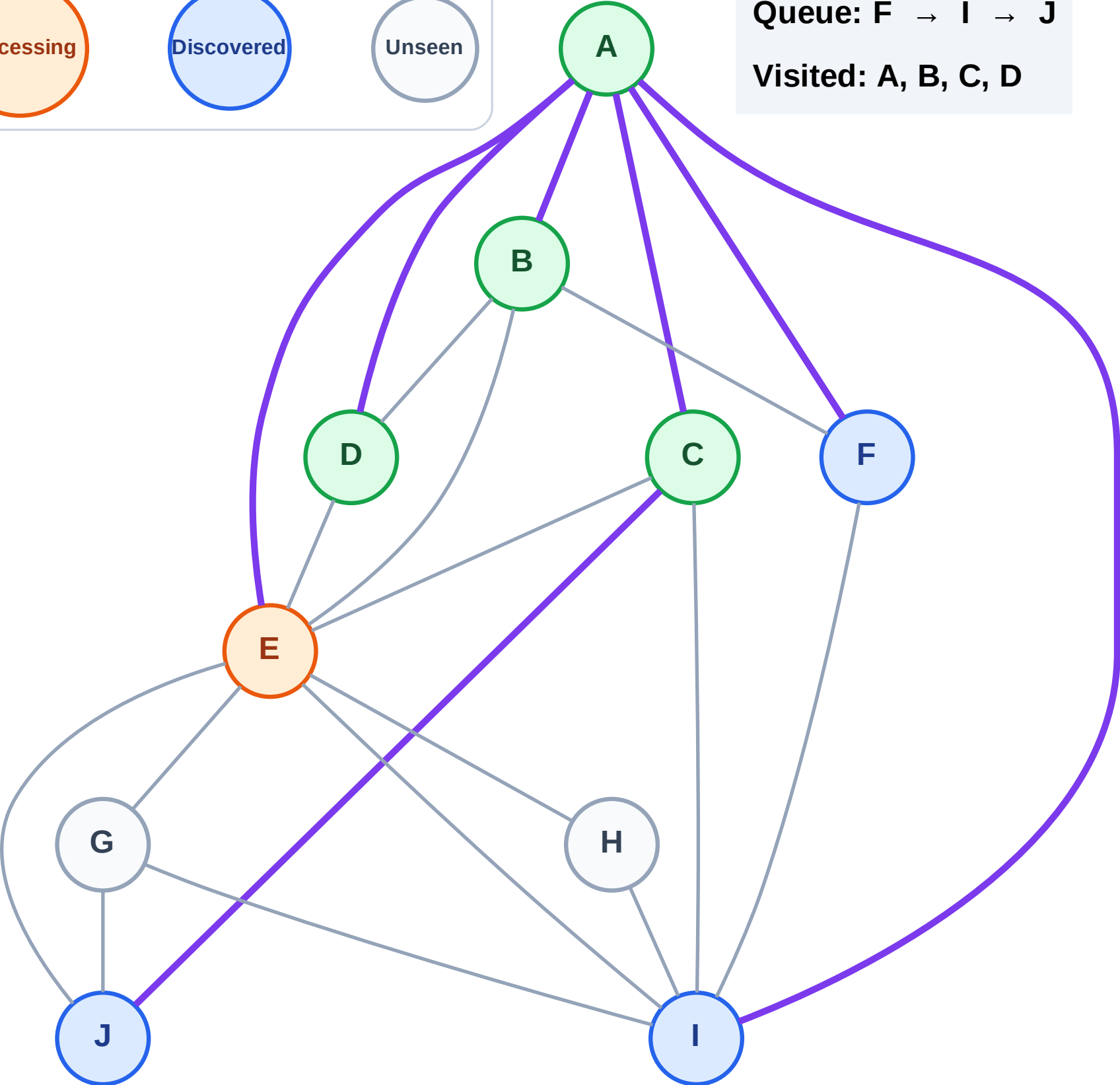
Complete

Processing

Discovered

Unseen

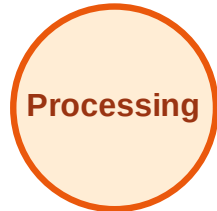
Queue: F → I → J
Visited: A, B, C, D



BFS Traversal

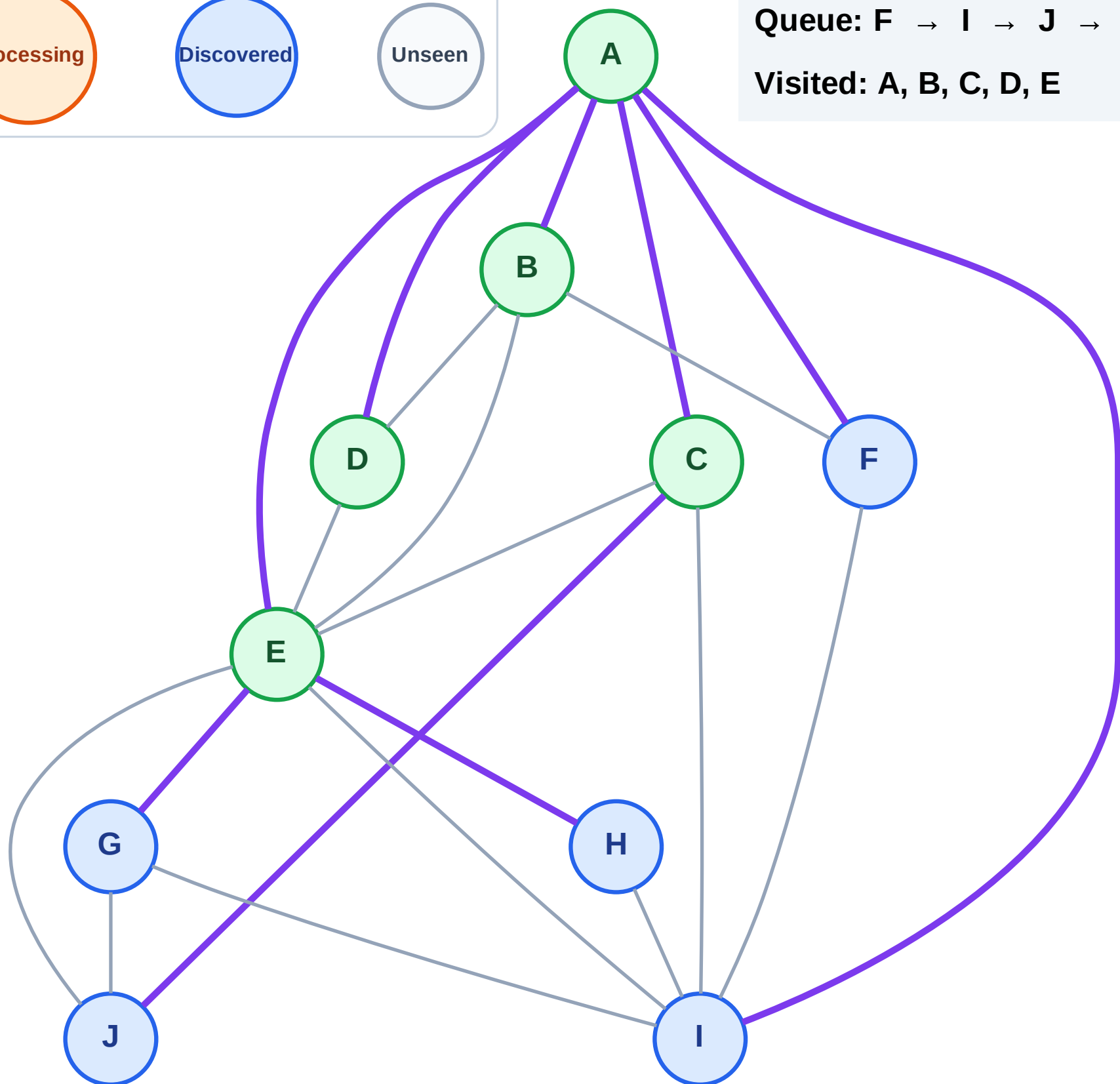
Step 11: Discover G, H from E

Legend



Queue: F → I → J → G → H

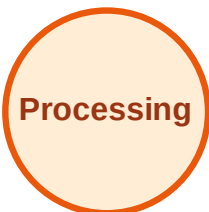
Visited: A, B, C, D, E



BFS Traversal

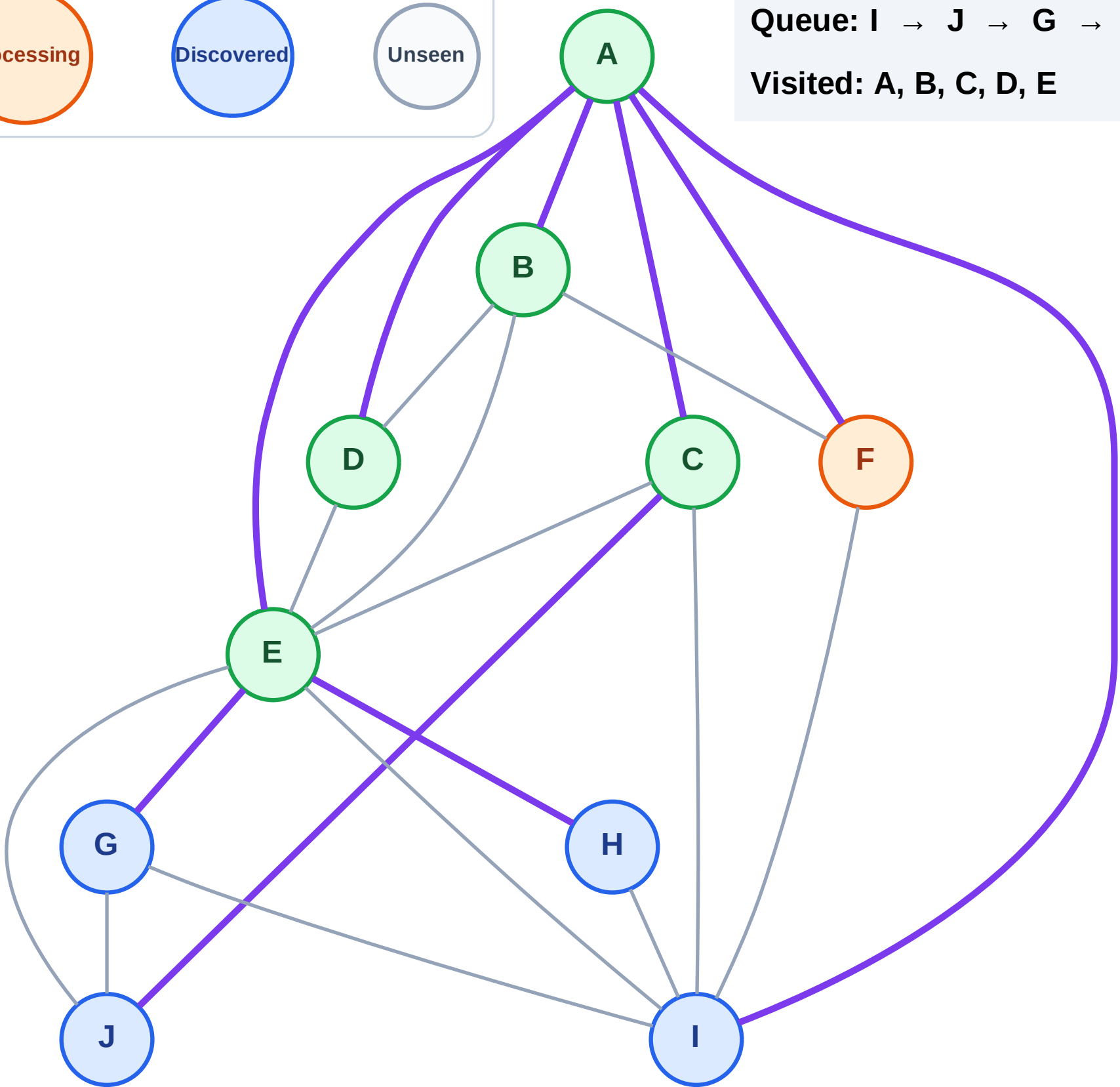
Step 12: Process node F

Legend



Queue: I → J → G → H

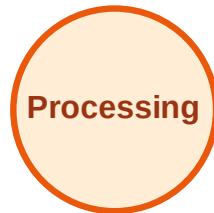
Visited: A, B, C, D, E



BFS Traversal

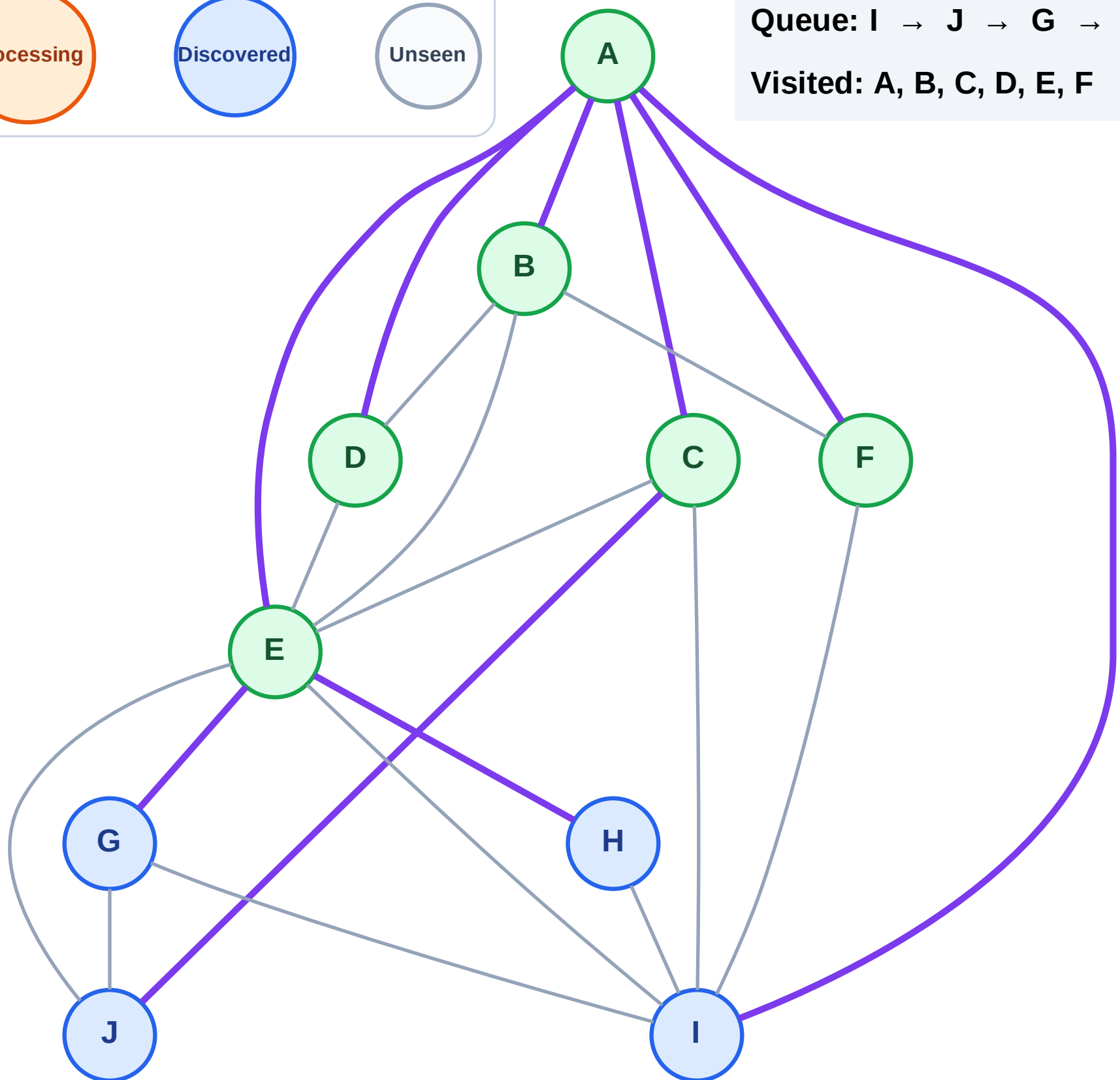
Step 13: No new neighbors from F

Legend



Queue: I → J → G → H

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node I

Legend

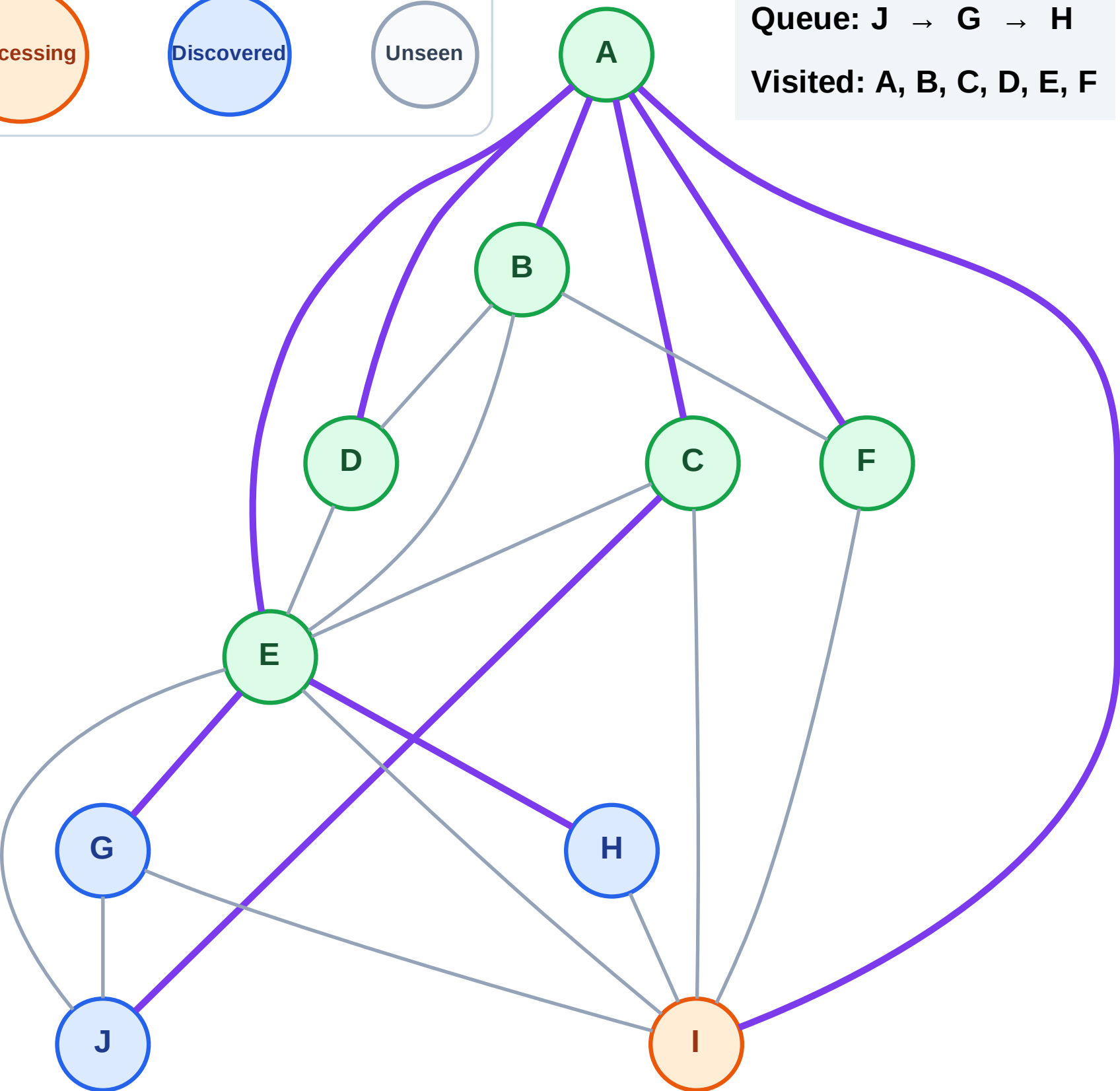
Complete

Processing

Discovered

Unseen

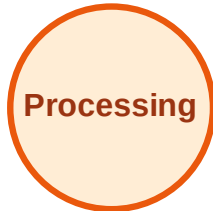
Queue: J → G → H
Visited: A, B, C, D, E, F



BFS Traversal

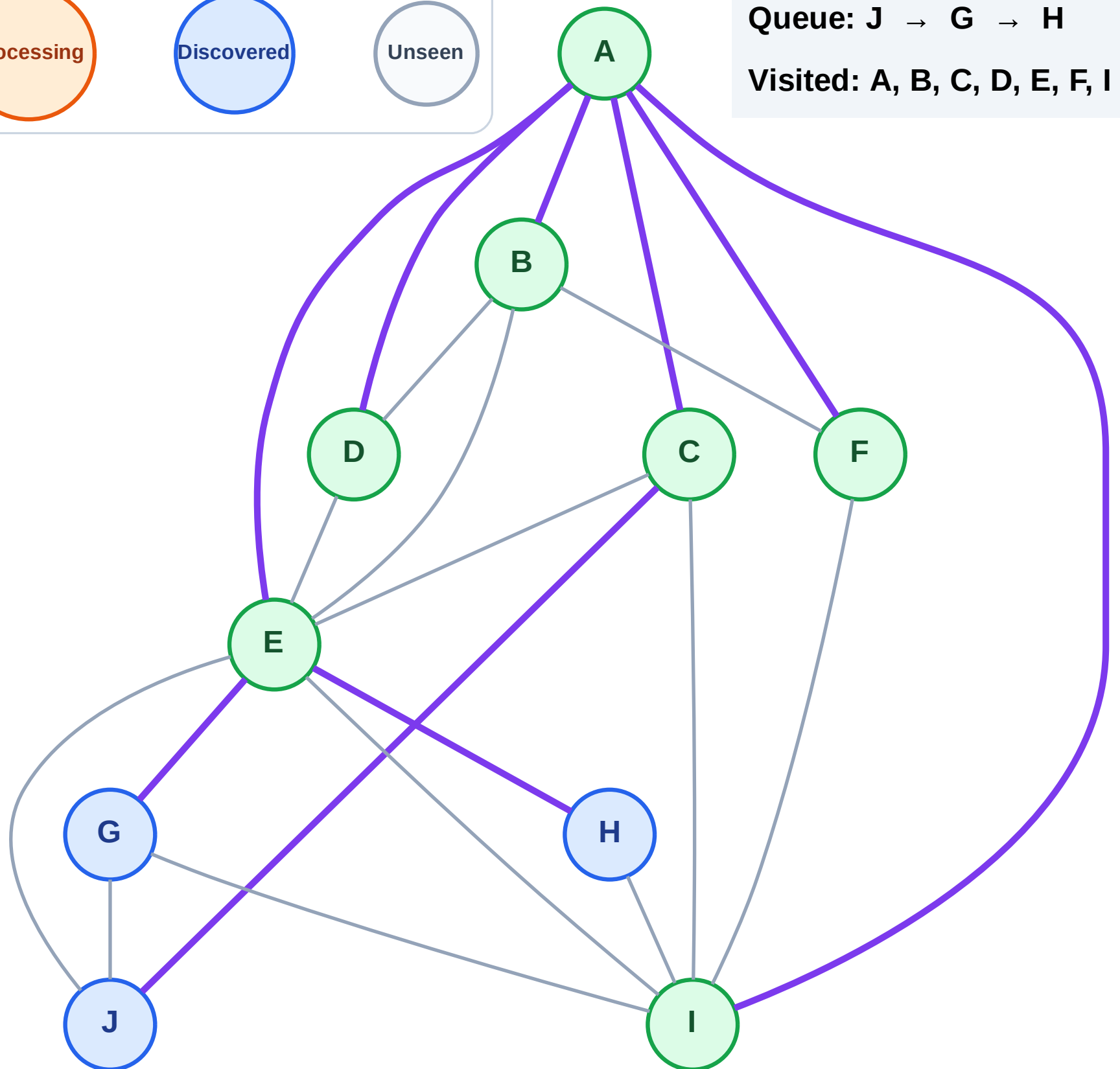
Step 15: No new neighbors from I

Legend



Queue: J → G → H

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node J

Legend

Complete

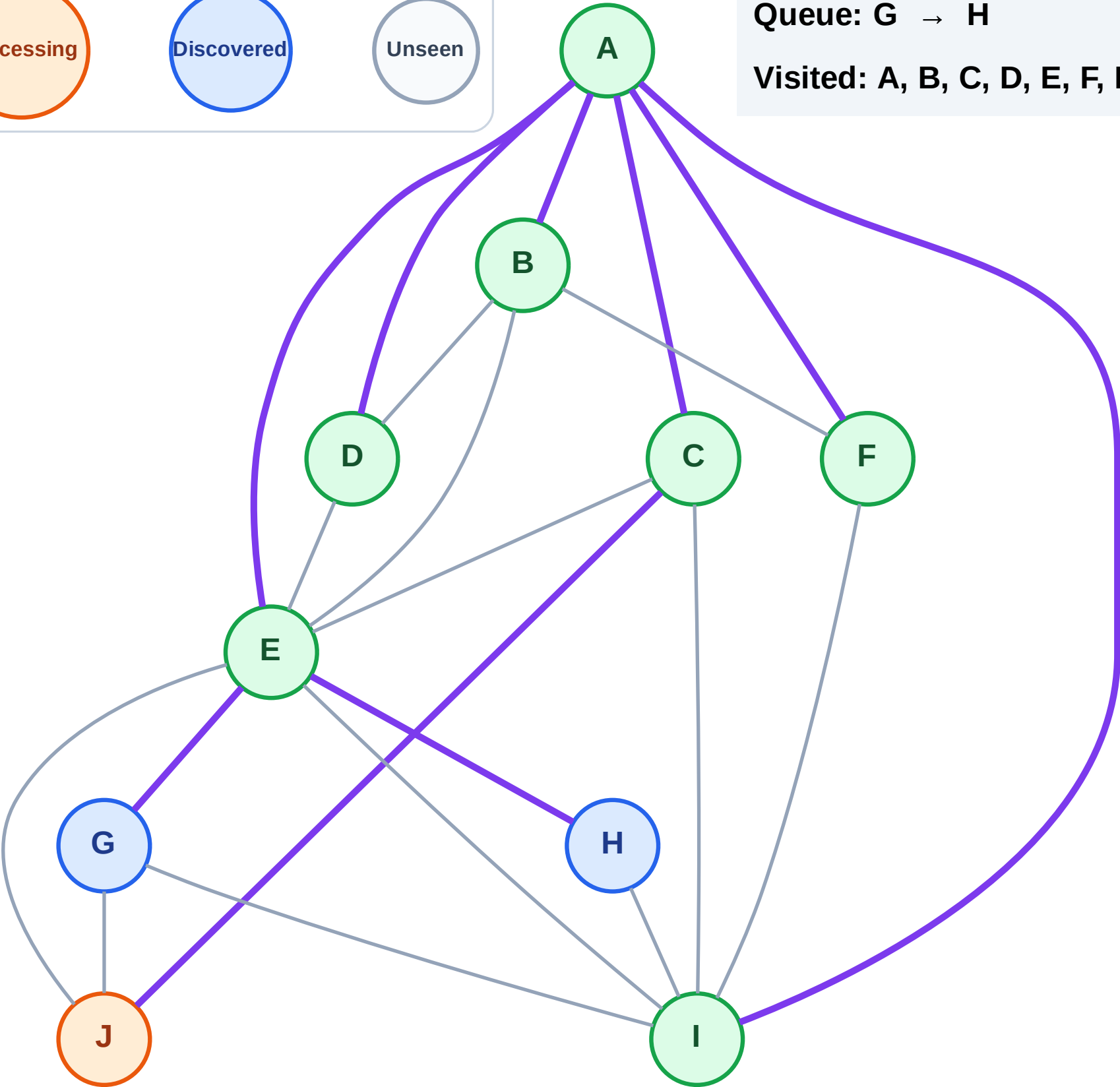
Processing

Discovered

Unseen

Queue: G → H

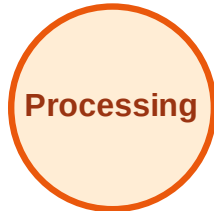
Visited: A, B, C, D, E, F, I



BFS Traversal

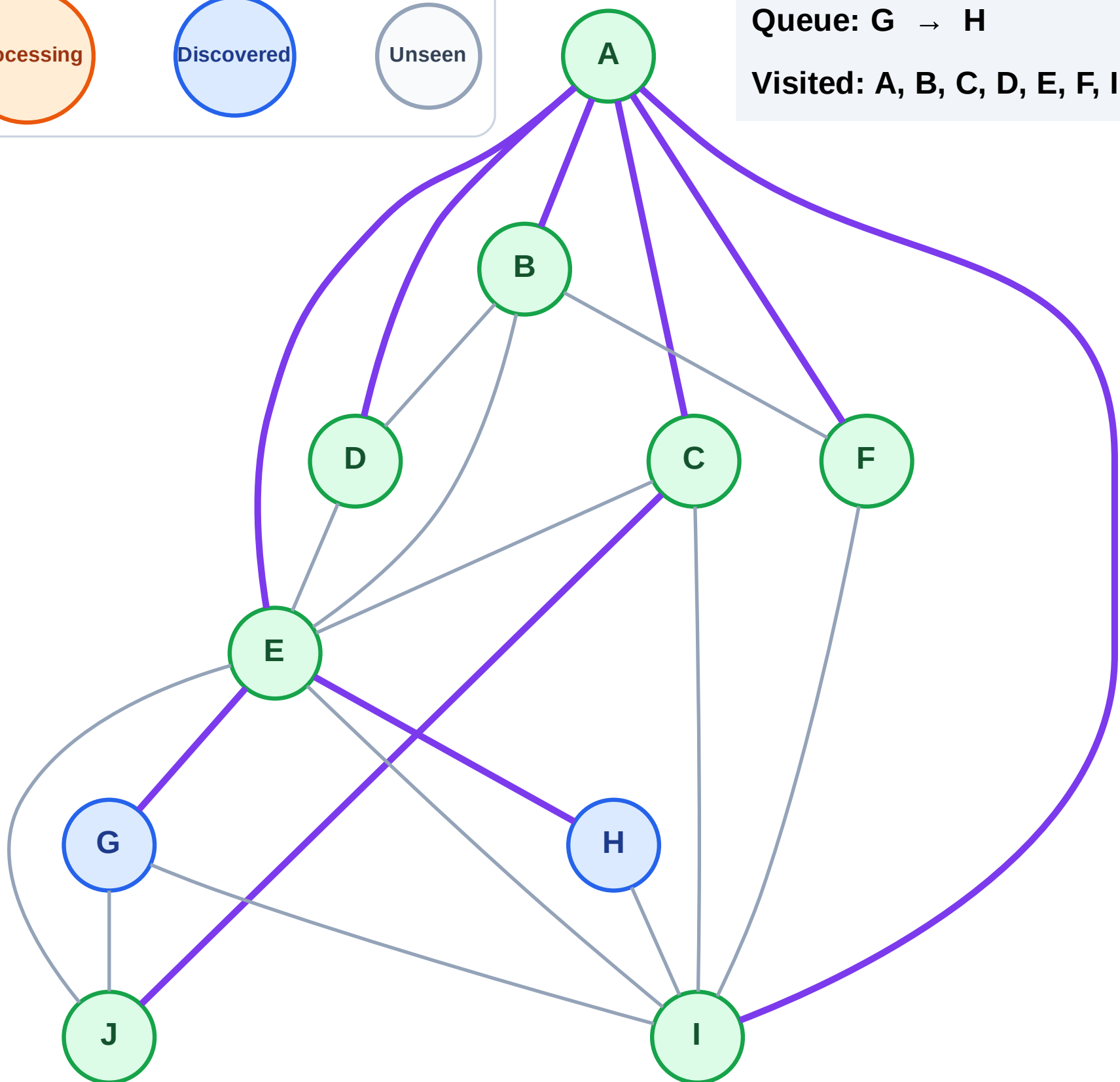
Step 17: No new neighbors from J

Legend



Queue: G → H

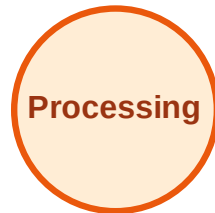
Visited: A, B, C, D, E, F, I, J



BFS Traversal

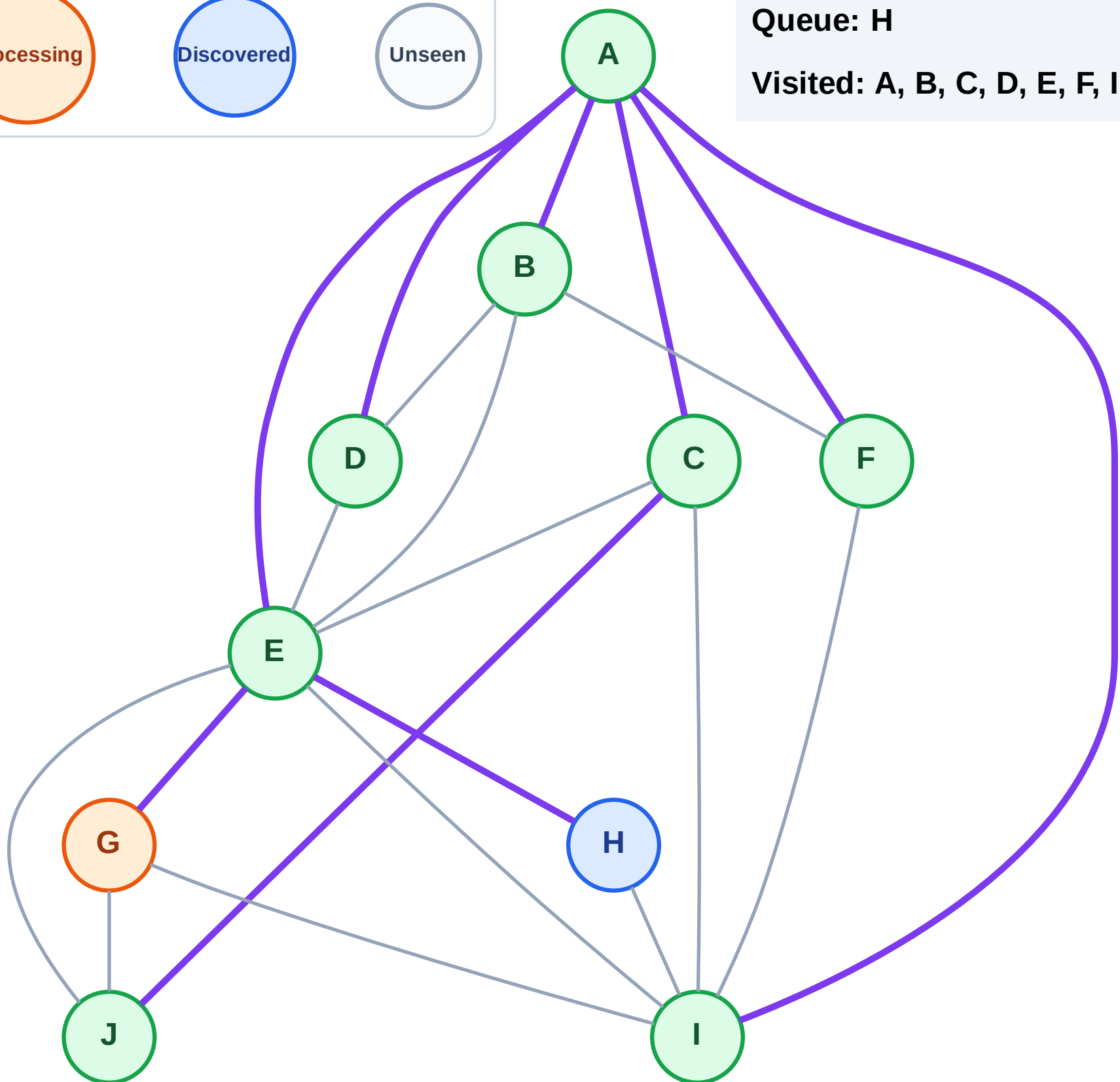
Step 18: Process node G

Legend



Queue: H

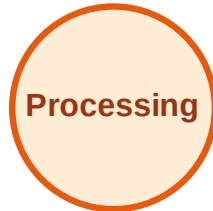
Visited: A, B, C, D, E, F, I, J



BFS Traversal

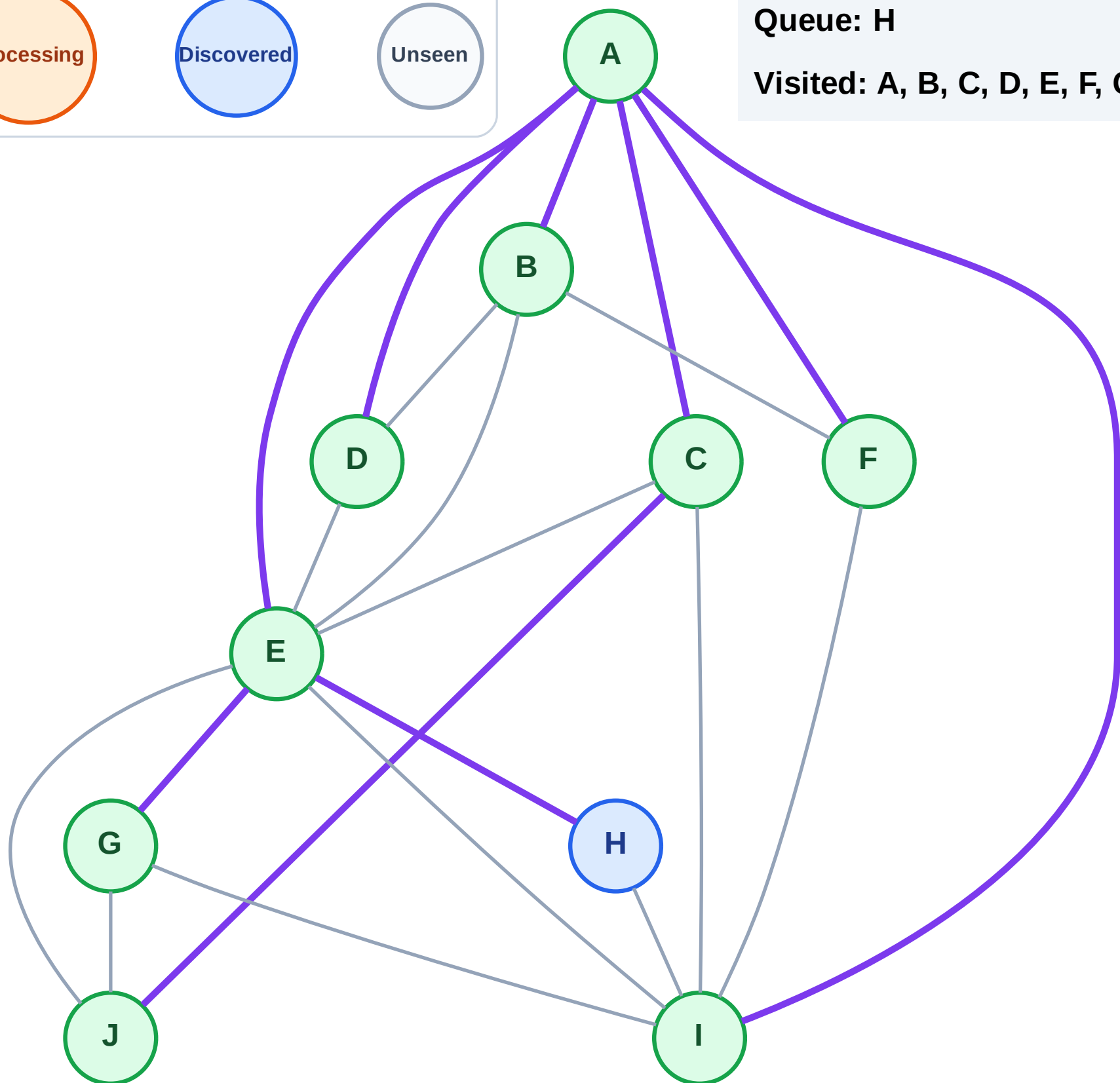
Step 19: No new neighbors from G

Legend



Queue: H

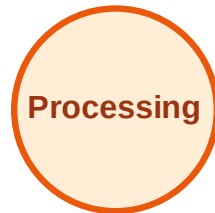
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

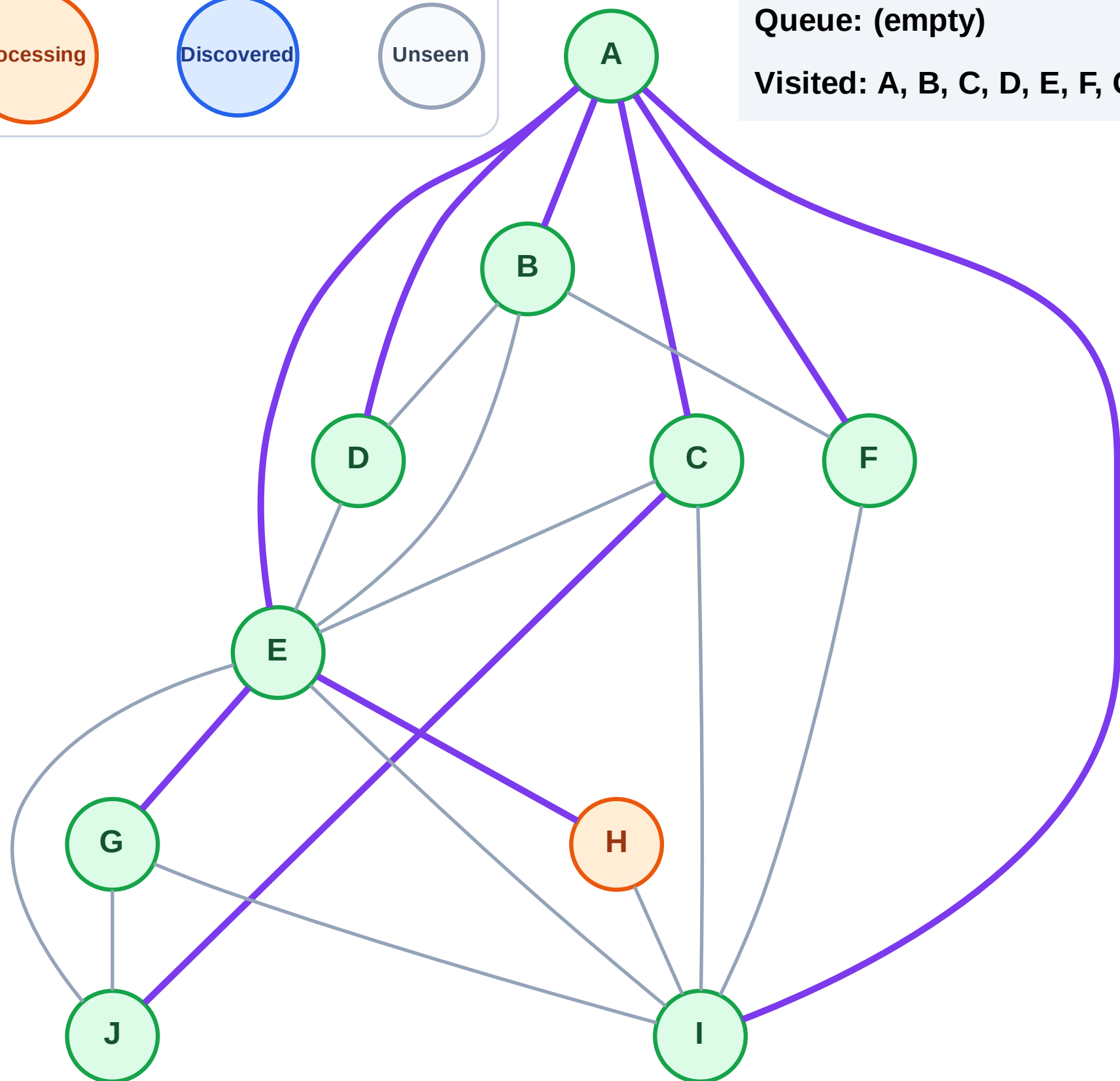
Step 20: Process node H

Legend



Queue: (empty)

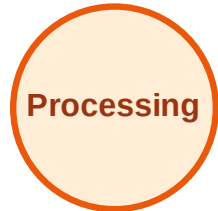
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

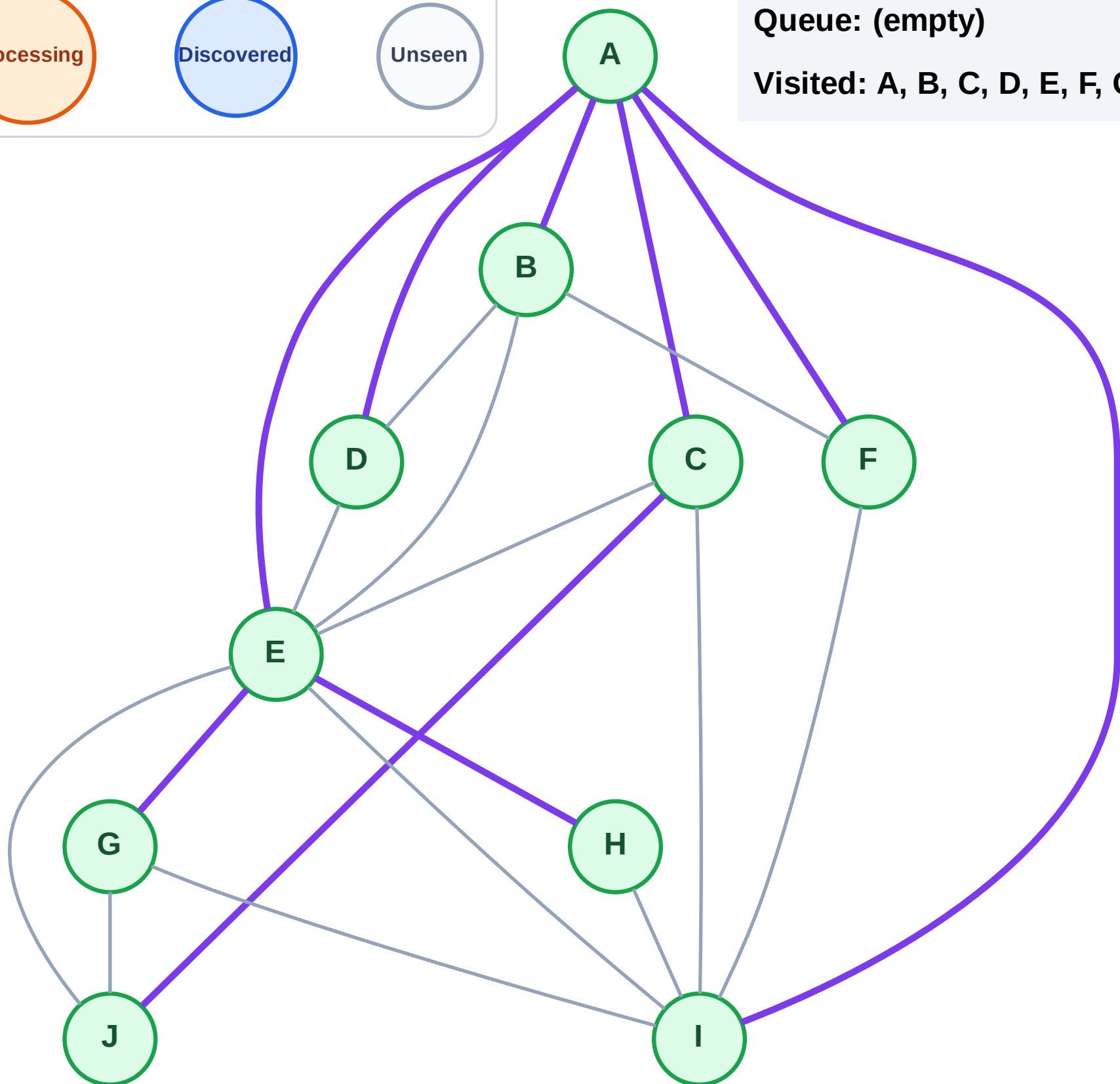
Step 21: No new neighbors from H

Legend



Queue: (empty)

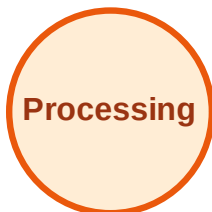
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

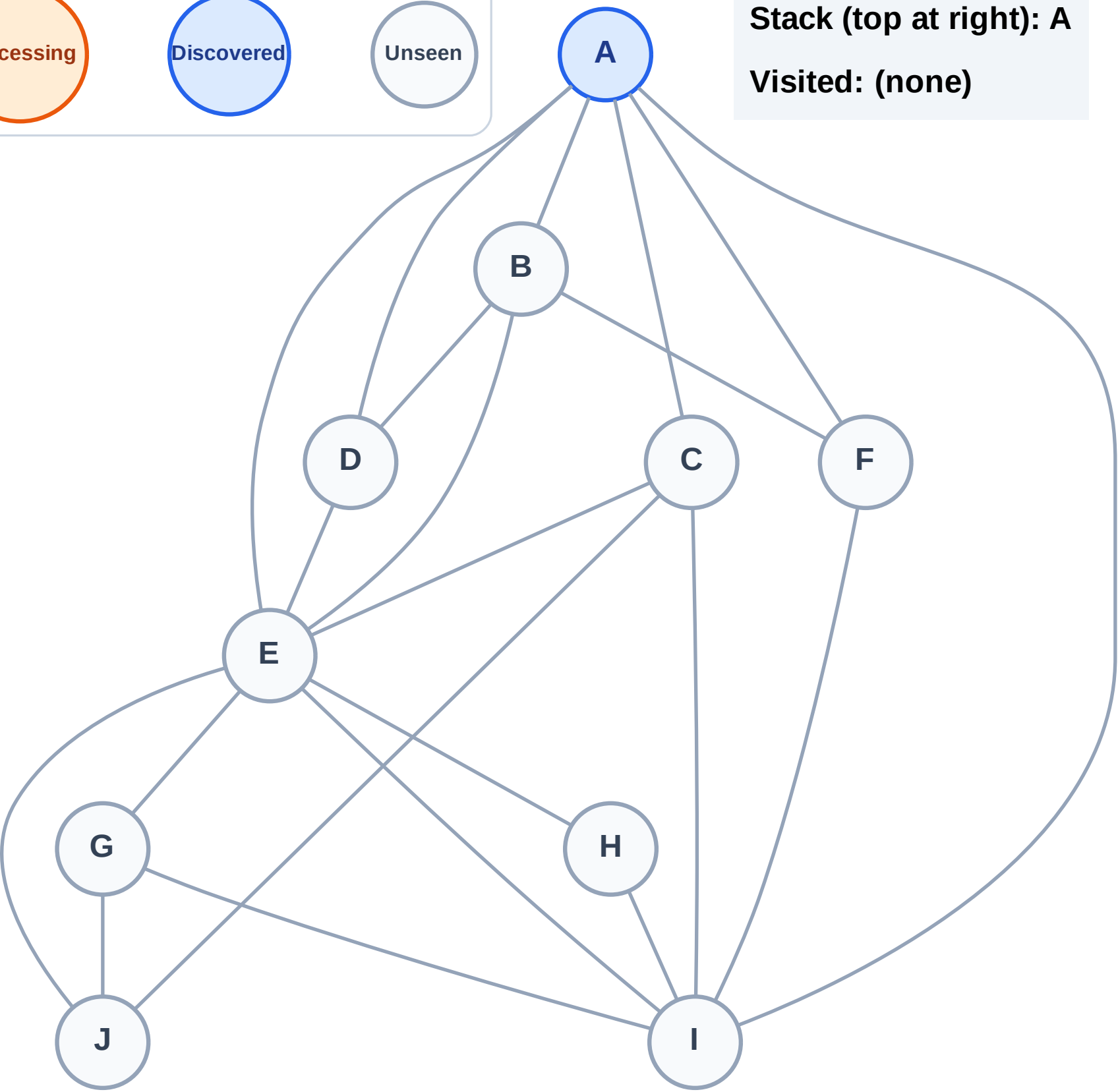
Step 1: Start at A

Legend



Stack (top at right): A

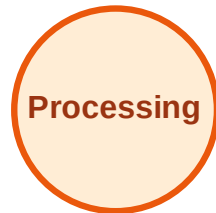
Visited: (none)



DFS Traversal

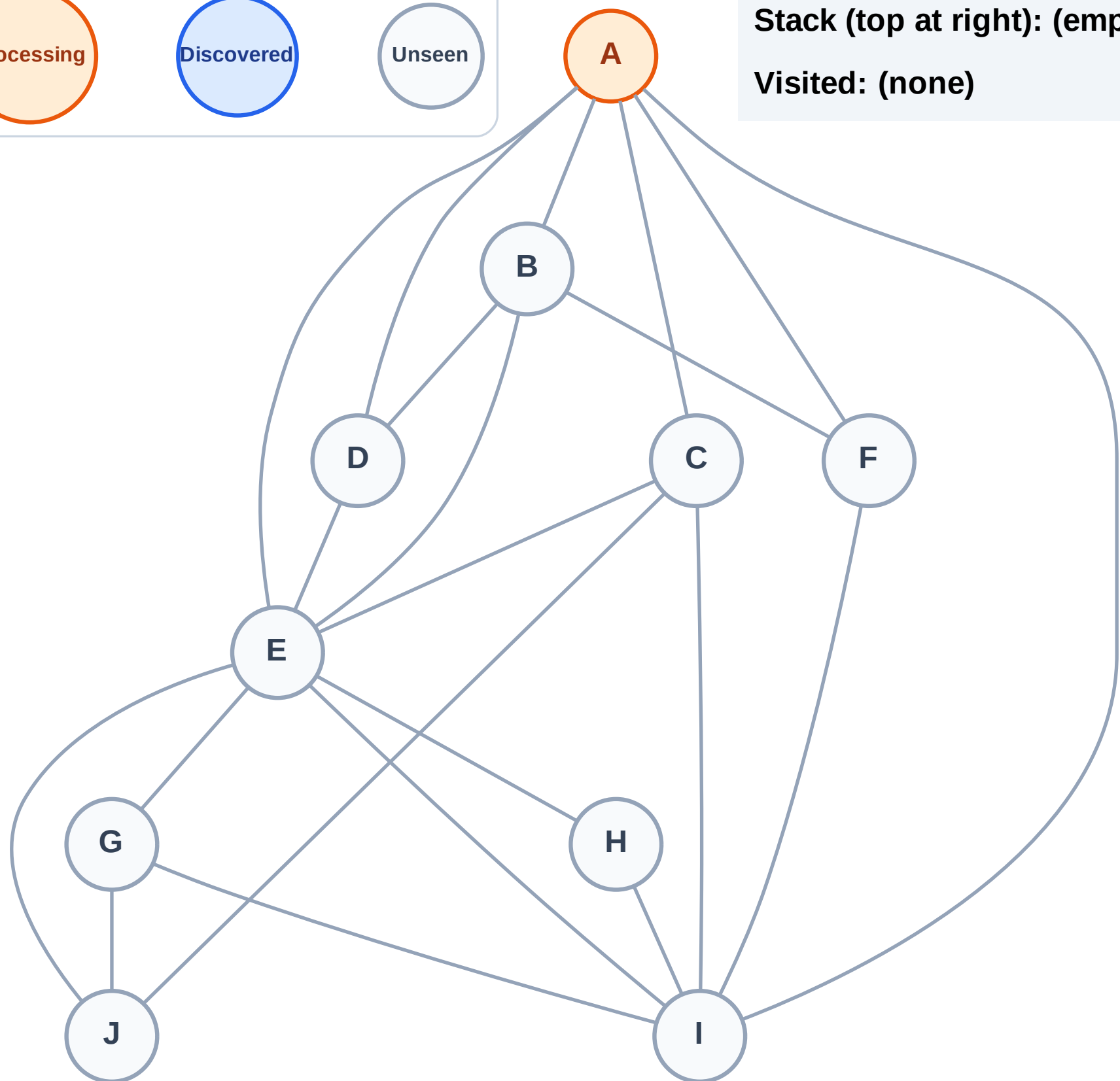
Step 2: Process node A

Legend



Stack (top at right): (empty)

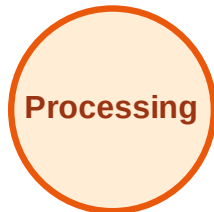
Visited: (none)



DFS Traversal

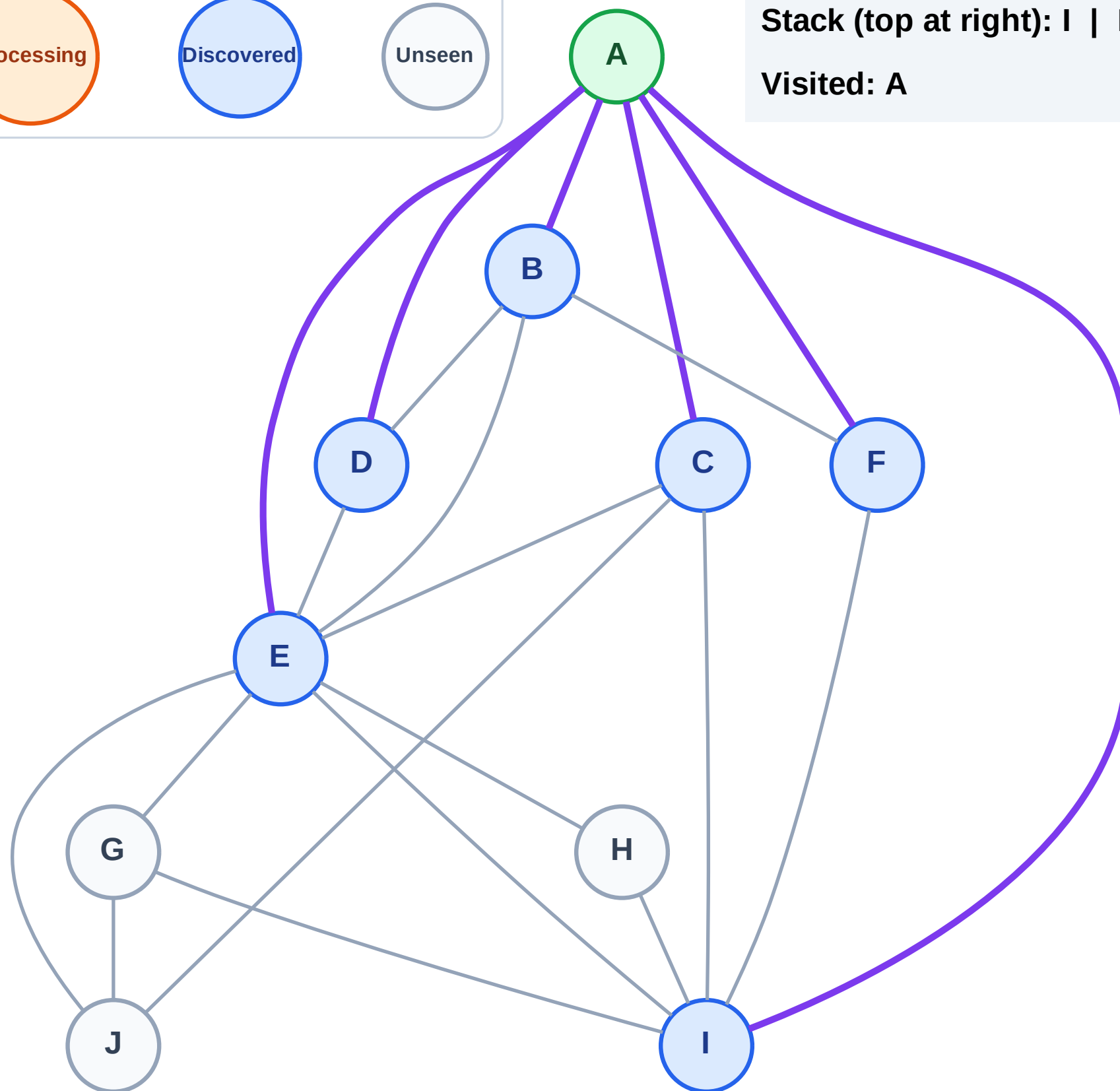
Step 3: Discover I, F, E, D, C, B from A

Legend



Stack (top at right): I | F | E | D | C | B

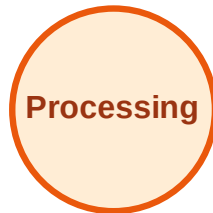
Visited: A



DFS Traversal

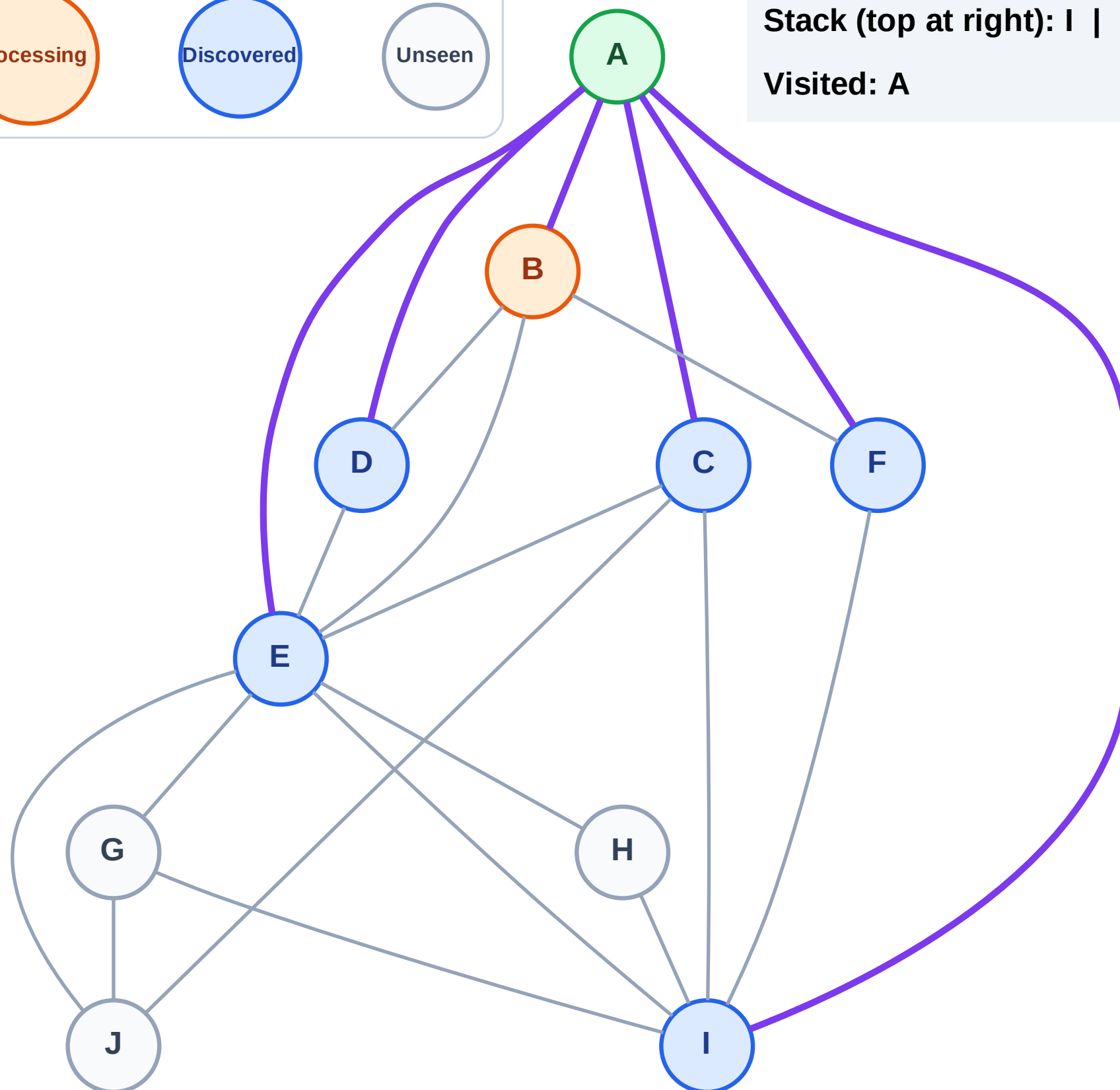
Step 4: Process node B

Legend



Stack (top at right): I | F | E | D | C

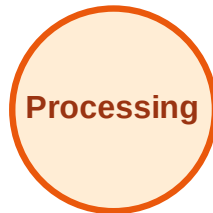
Visited: A



DFS Traversal

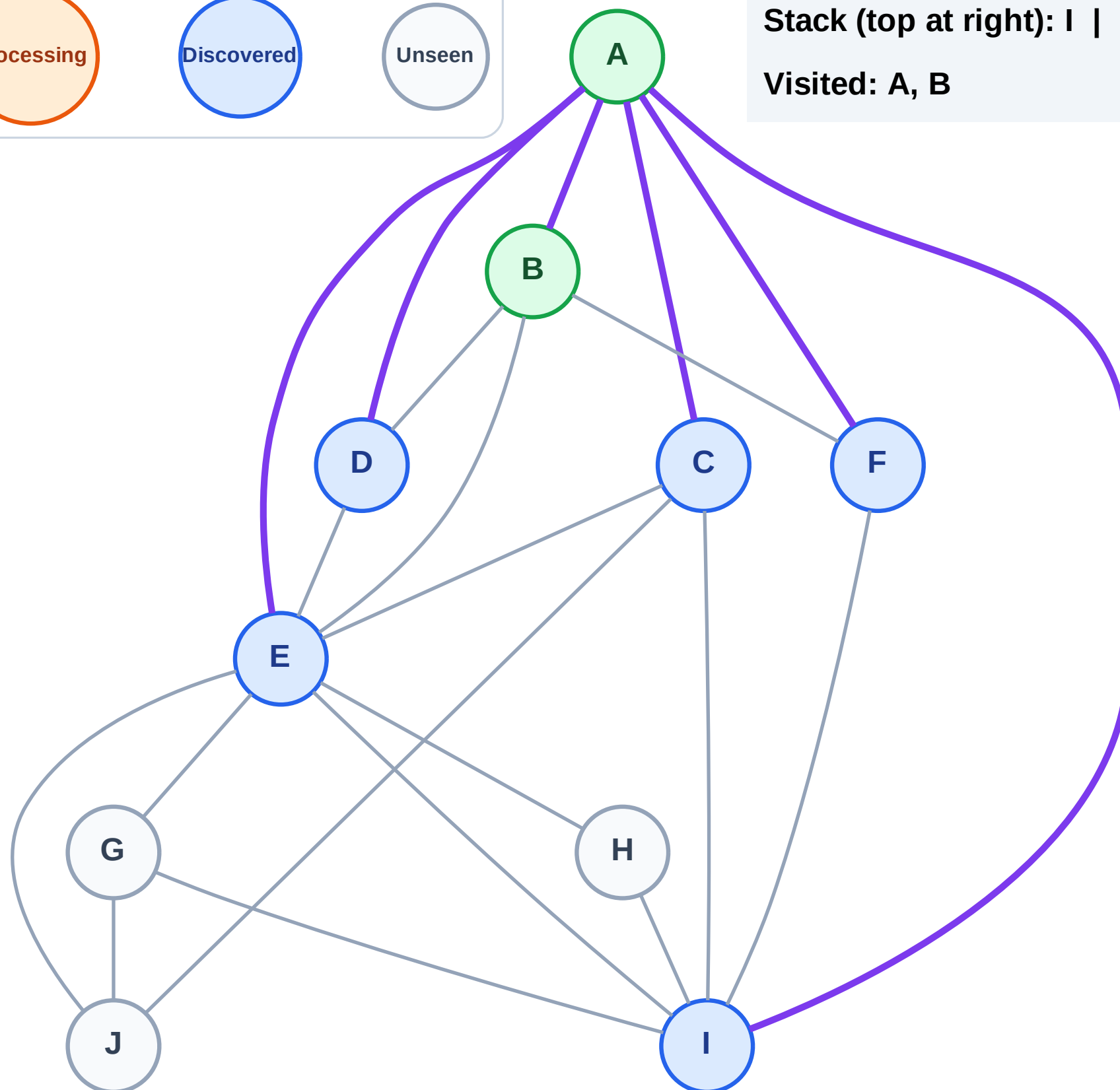
Step 5: No new neighbors from B

Legend



Stack (top at right): I | F | E | D | C

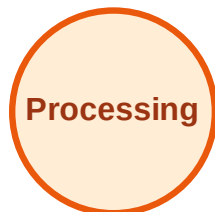
Visited: A, B



DFS Traversal

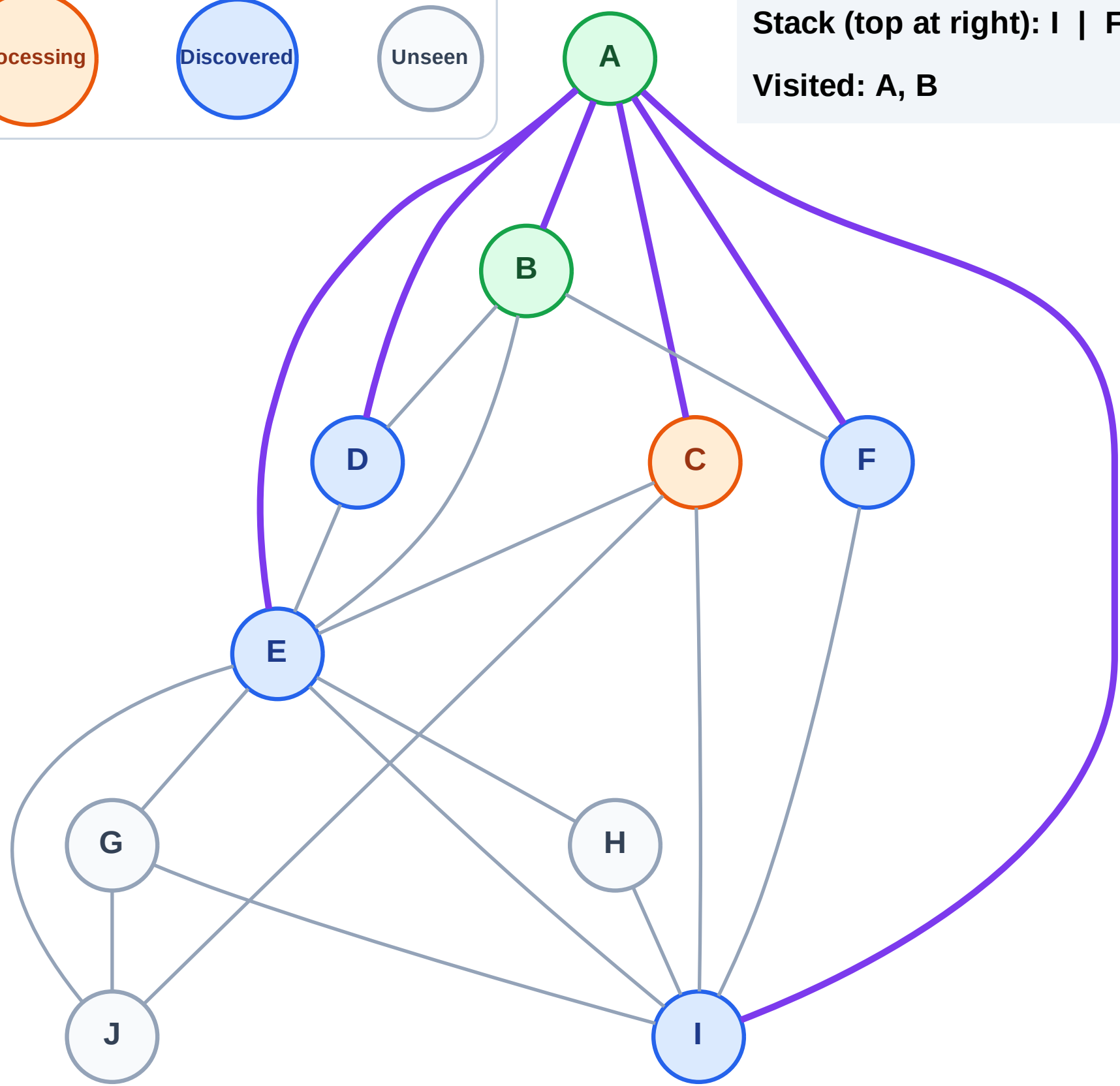
Step 6: Process node C

Legend



Stack (top at right): I | F | E | D

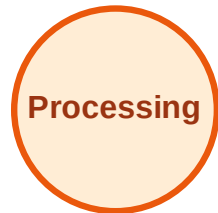
Visited: A, B



DFS Traversal

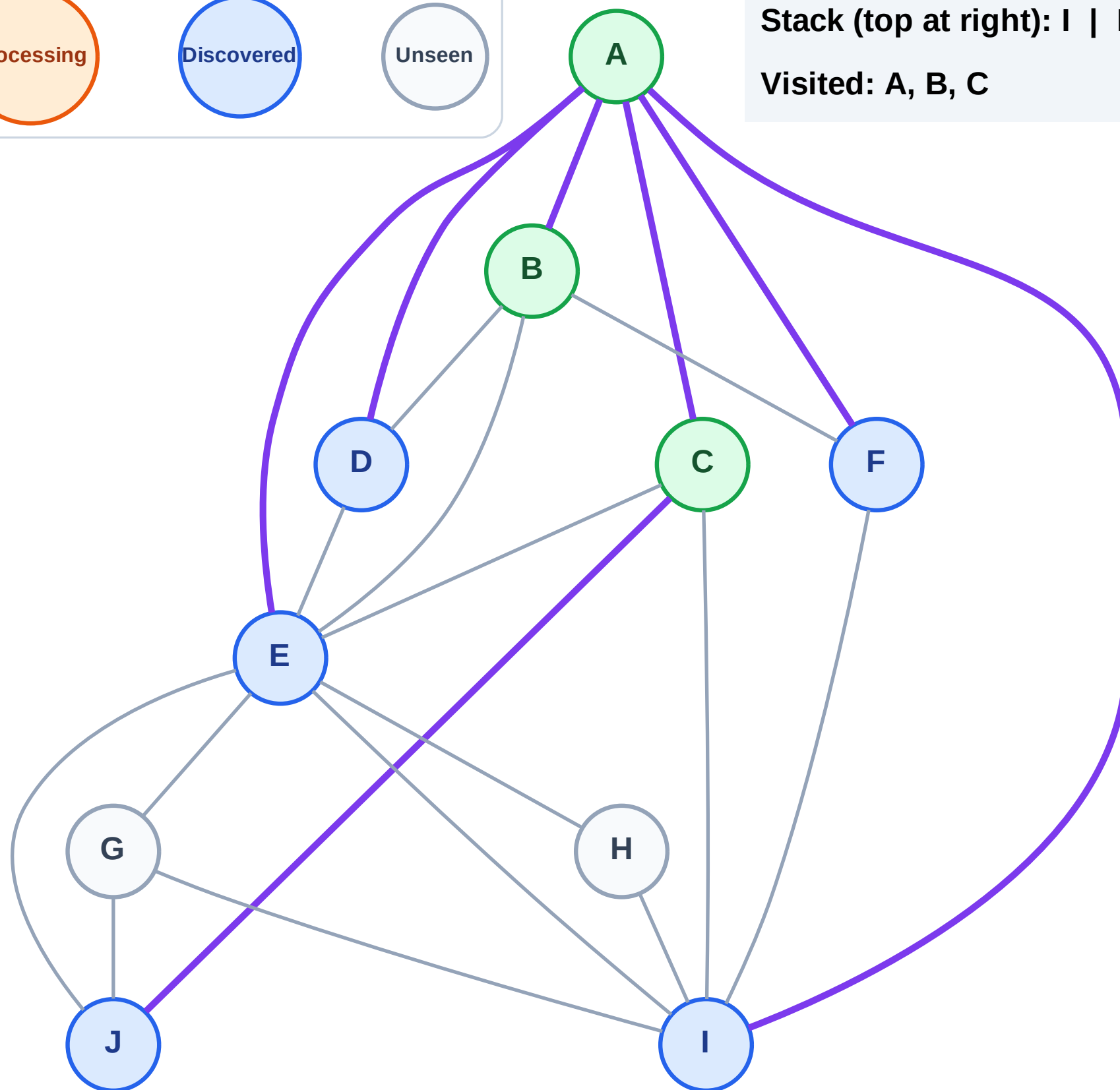
Step 7: Discover J from C

Legend



Stack (top at right): I | F | E | D | J

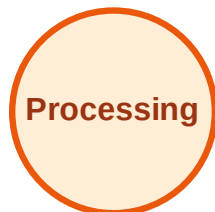
Visited: A, B, C



DFS Traversal

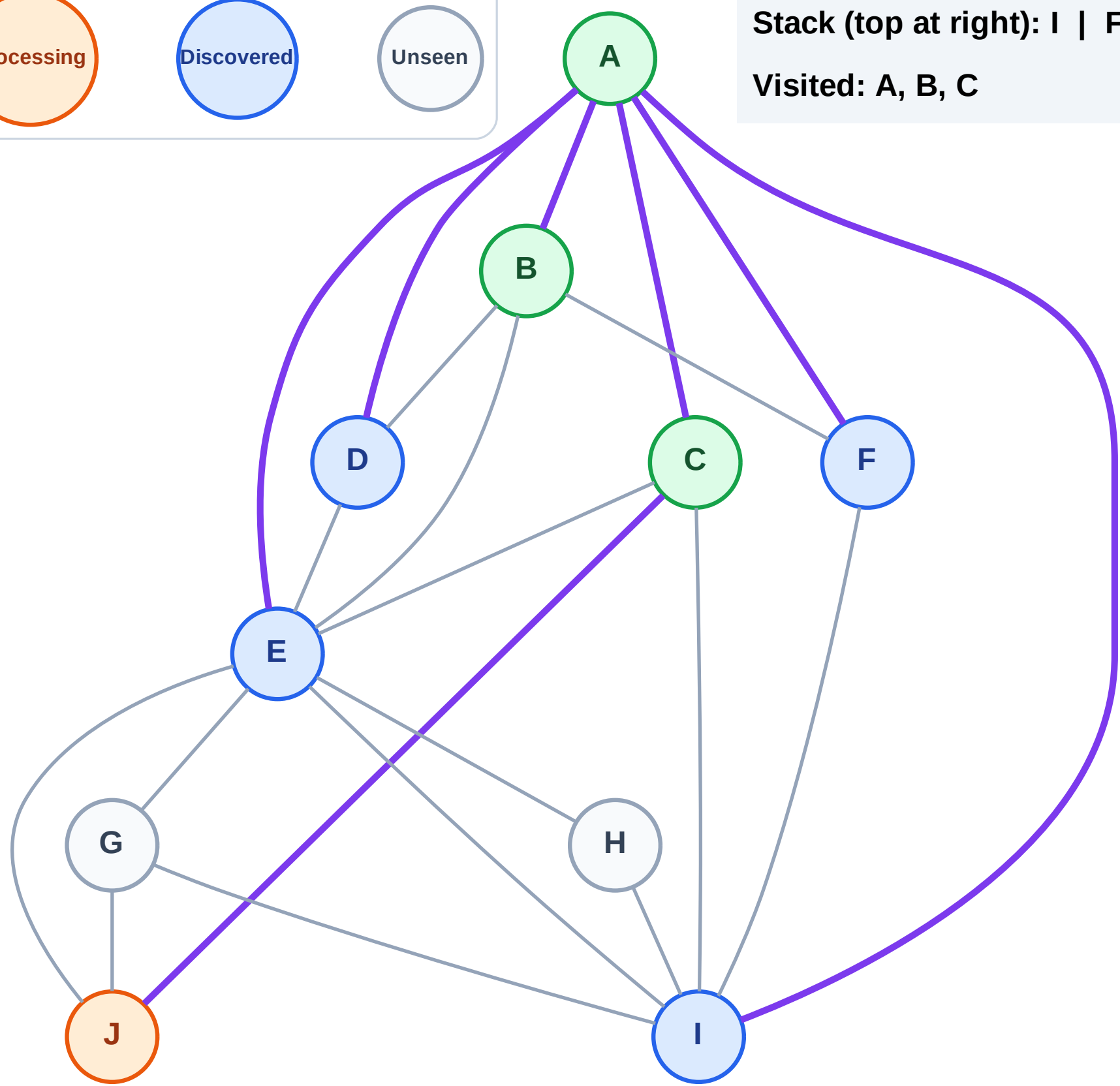
Step 8: Process node J

Legend



Stack (top at right): I | F | E | D

Visited: A, B, C

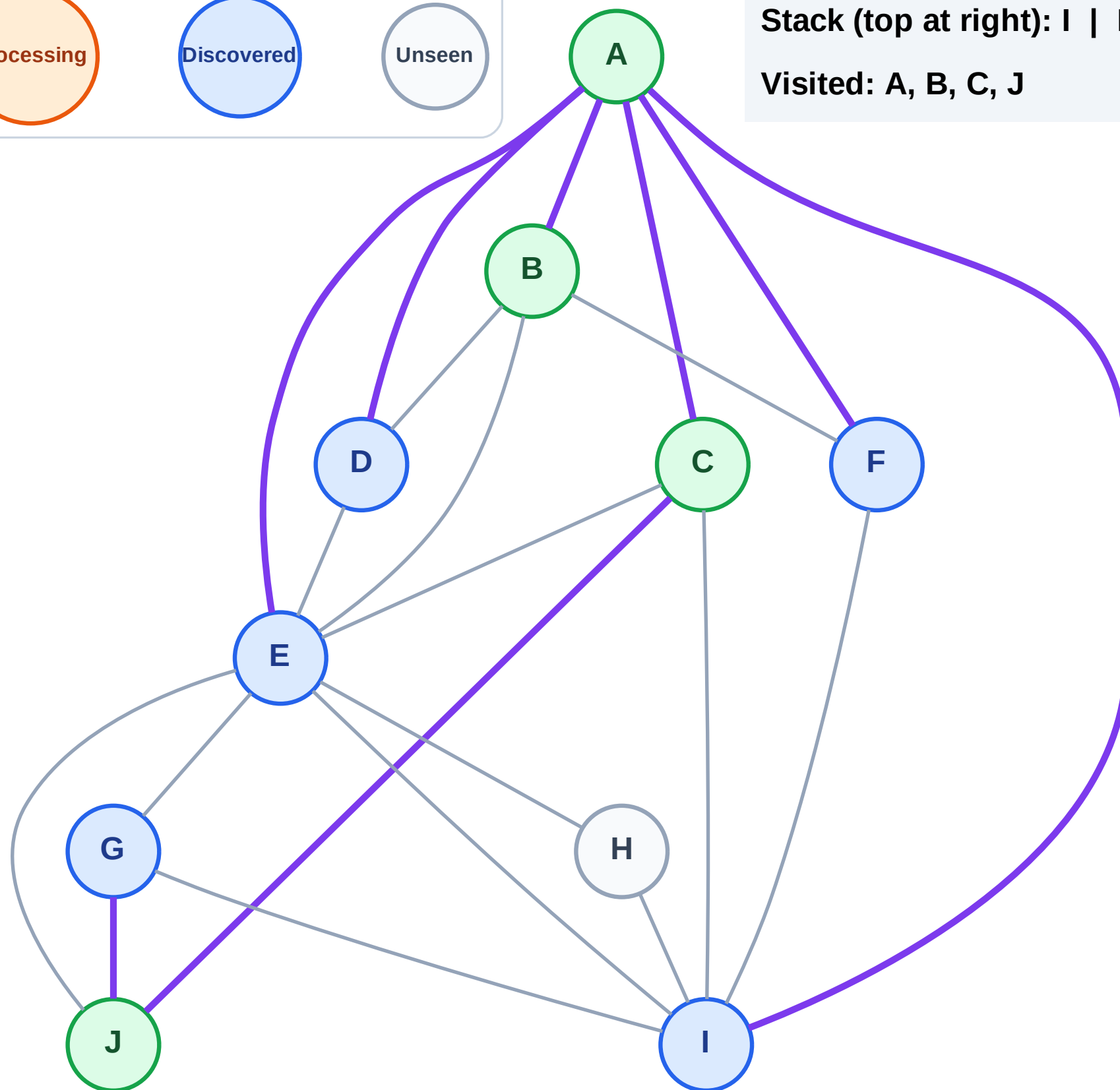


Step 9: Discover G from J

Legend



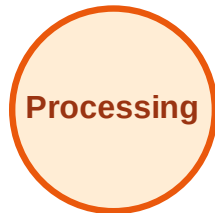
Visited: A, B, C, J



DFS Traversal

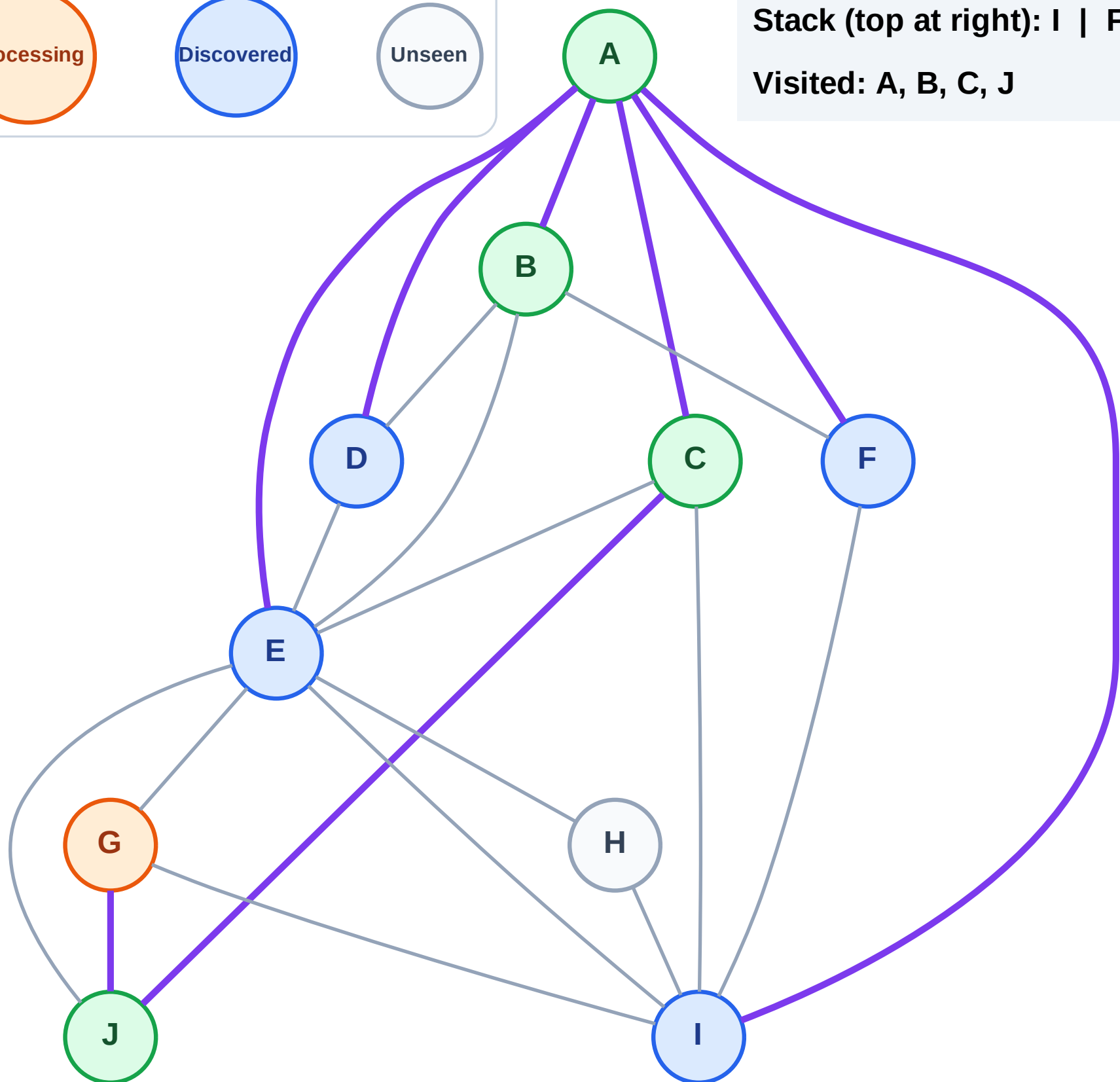
Step 10: Process node G

Legend



Stack (top at right): I | F | E | D

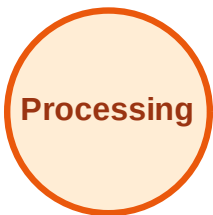
Visited: A, B, C, J



DFS Traversal

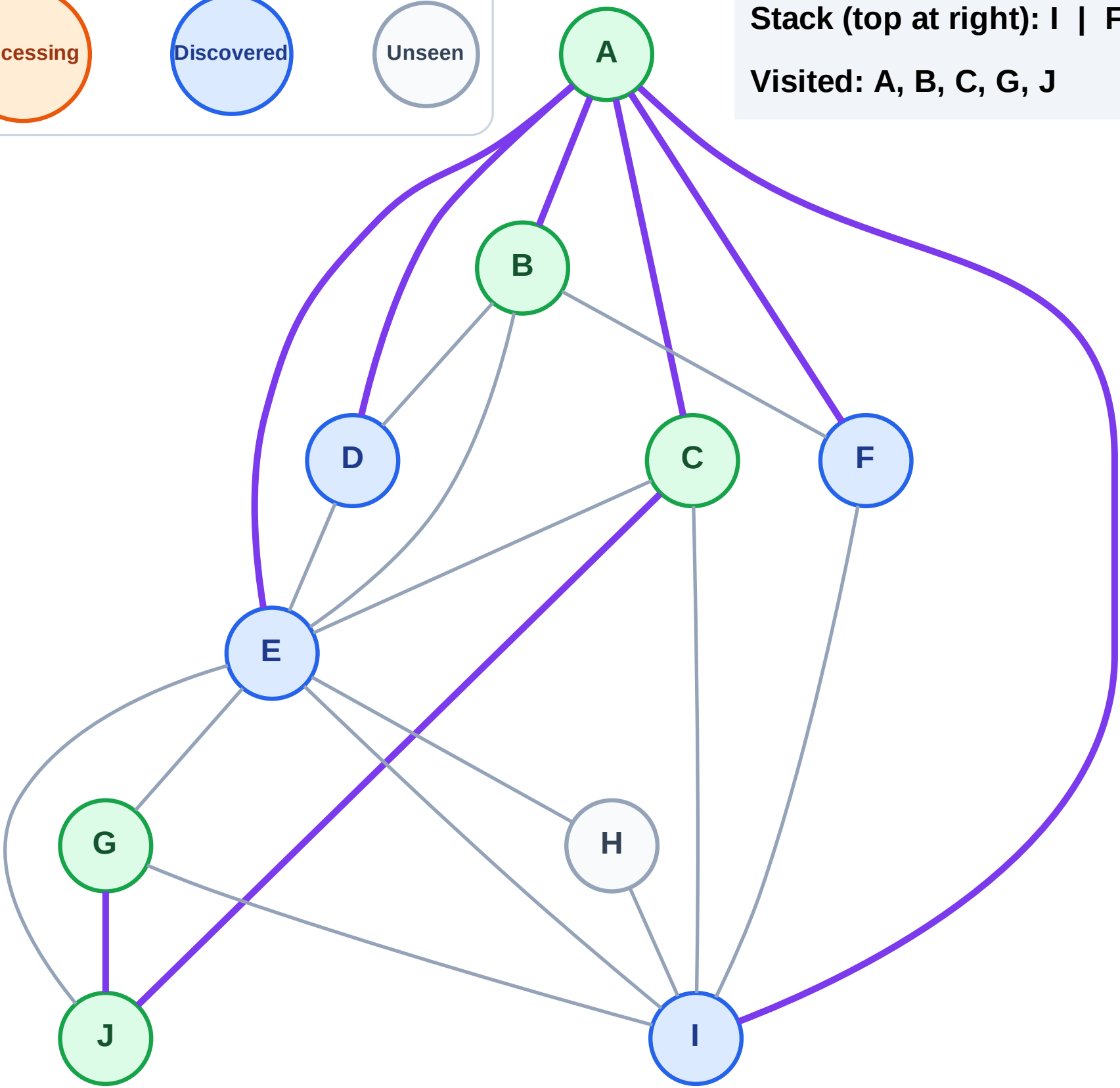
Step 11: No new neighbors from G

Legend



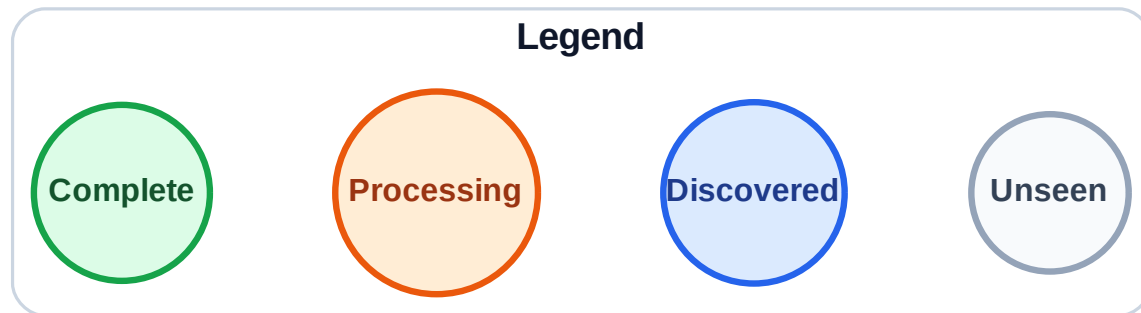
Stack (top at right): I | F | E | D

Visited: A, B, C, G, J



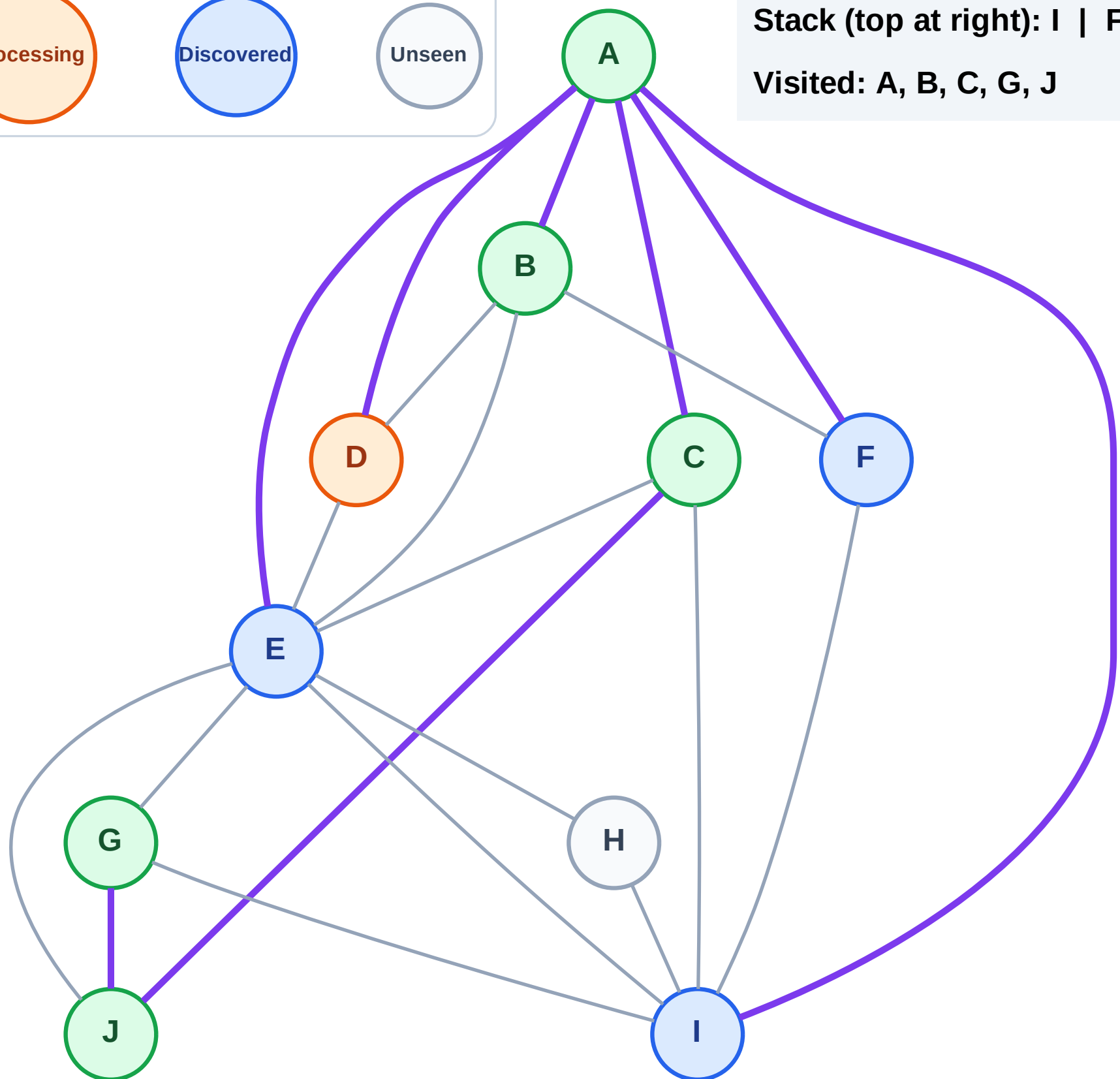
Step 12: Process node D

Step 12: Process node D



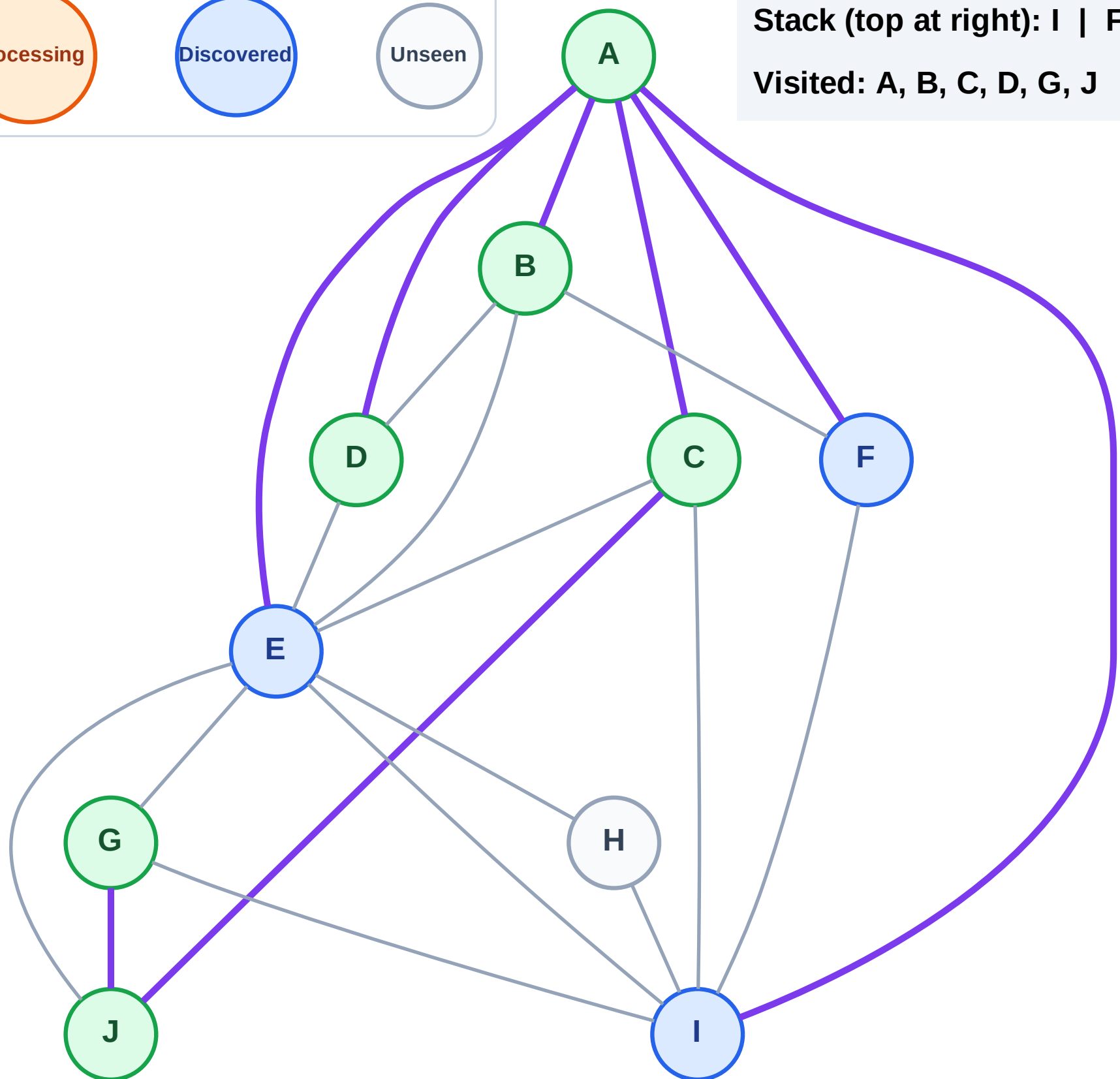
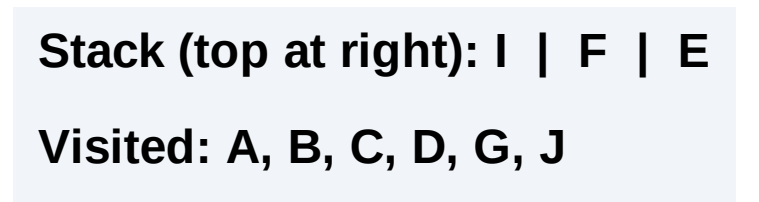
Stack (top at right): I | F | E

Visited: A, B, C, G, J



Step 13: No new neighbors from D

Step 13: No new neighbors from D



DFS Traversal

Step 14: Process node E

Legend

Complete

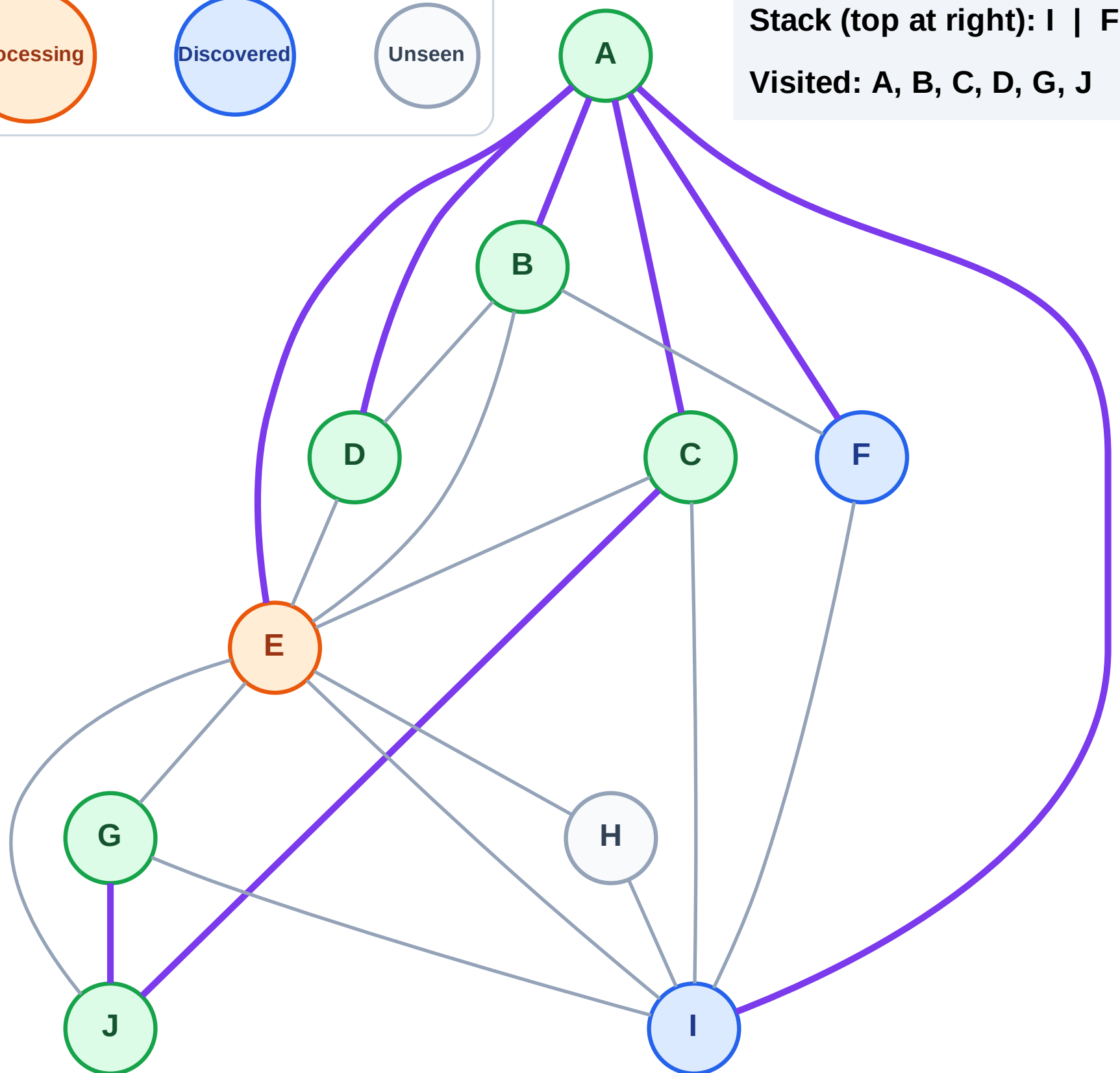
Processing

Discovered

Unseen

Stack (top at right): I | F

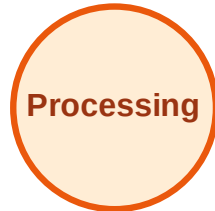
Visited: A, B, C, D, G, J



DFS Traversal

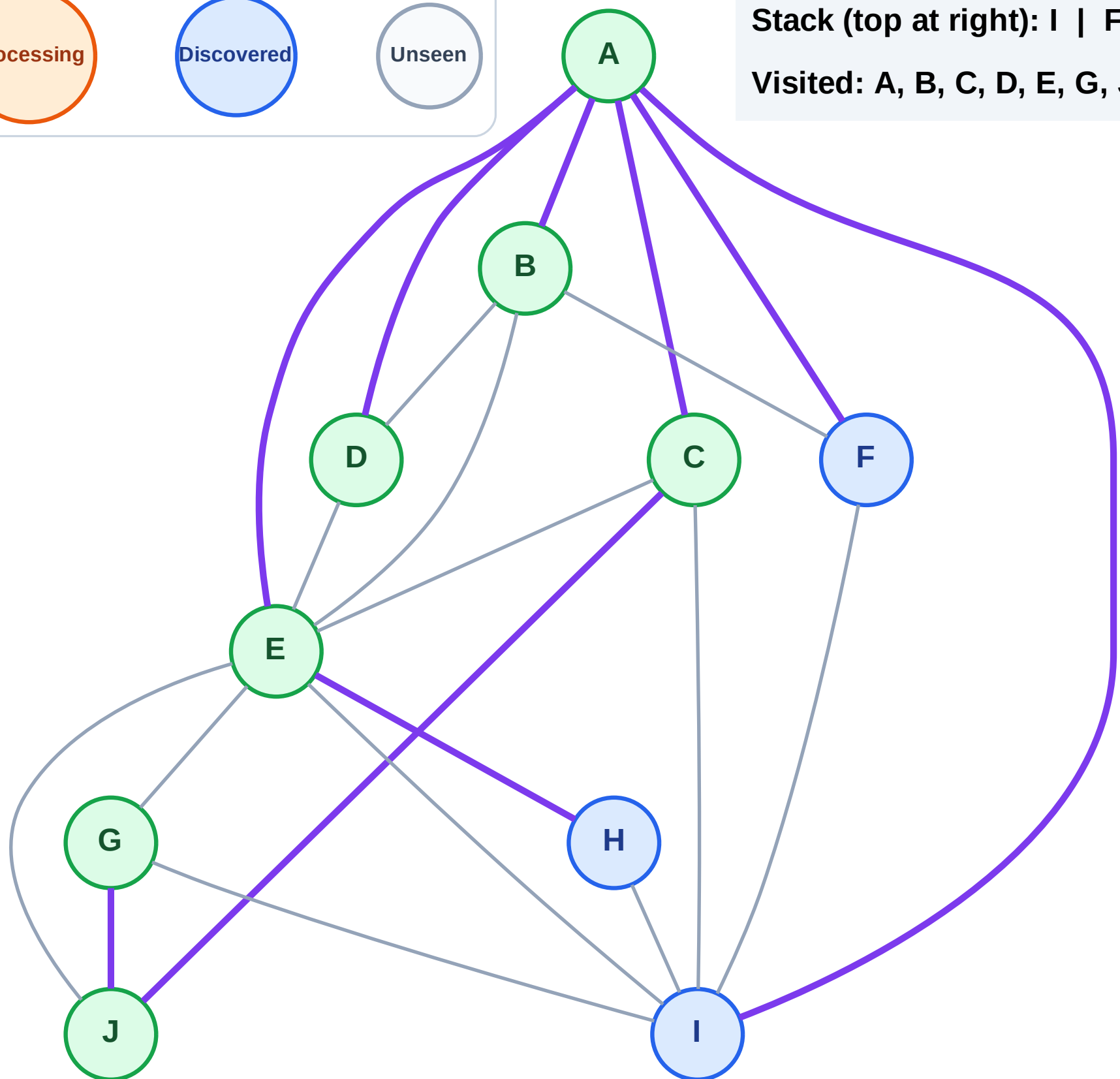
Step 15: Discover H from E

Legend



Stack (top at right): I | F | H

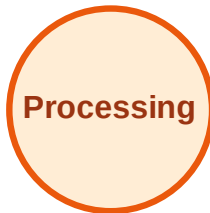
Visited: A, B, C, D, E, G, J



DFS Traversal

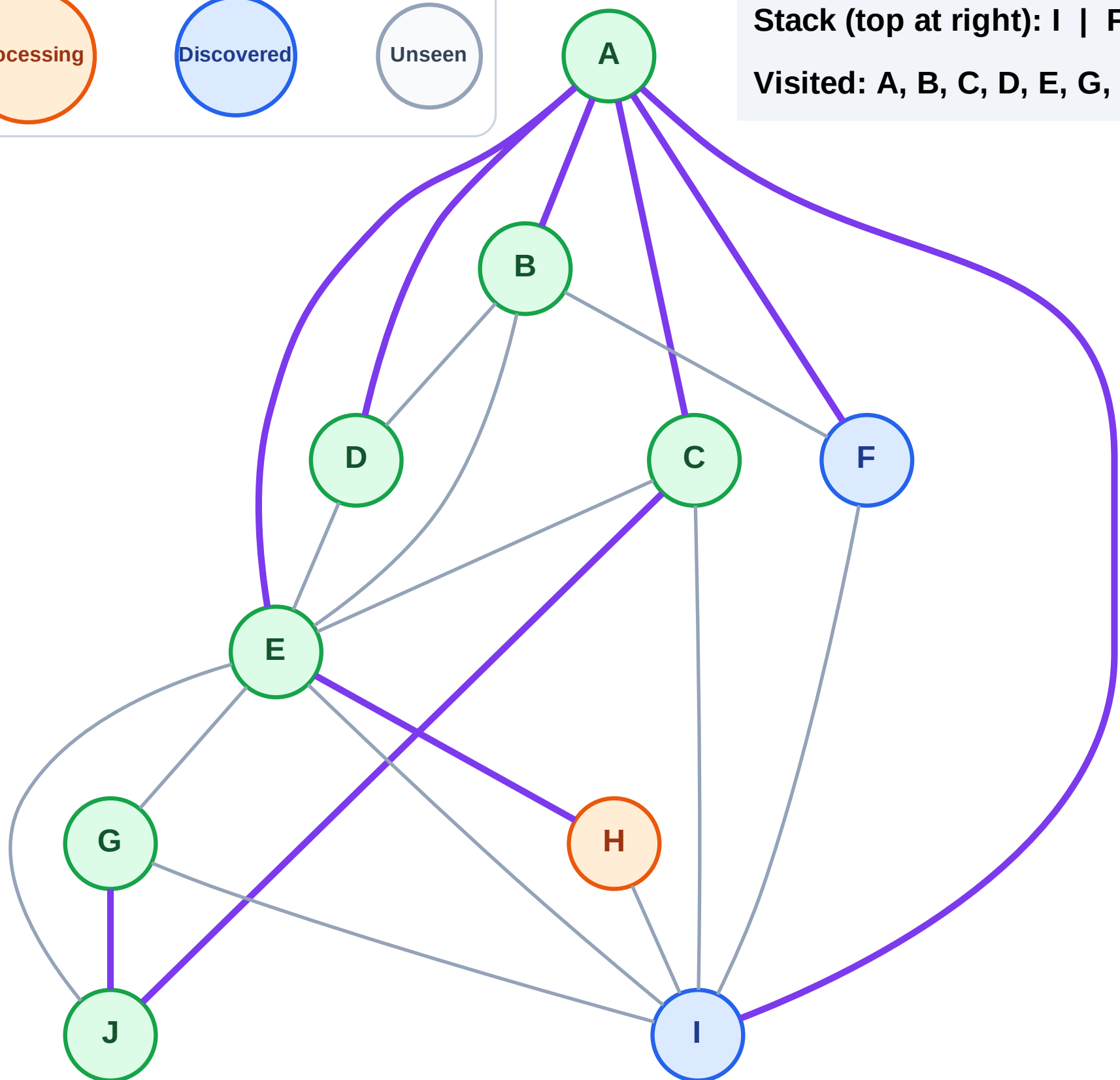
Step 16: Process node H

Legend



Stack (top at right): I | F

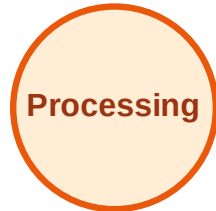
Visited: A, B, C, D, E, G, J



DFS Traversal

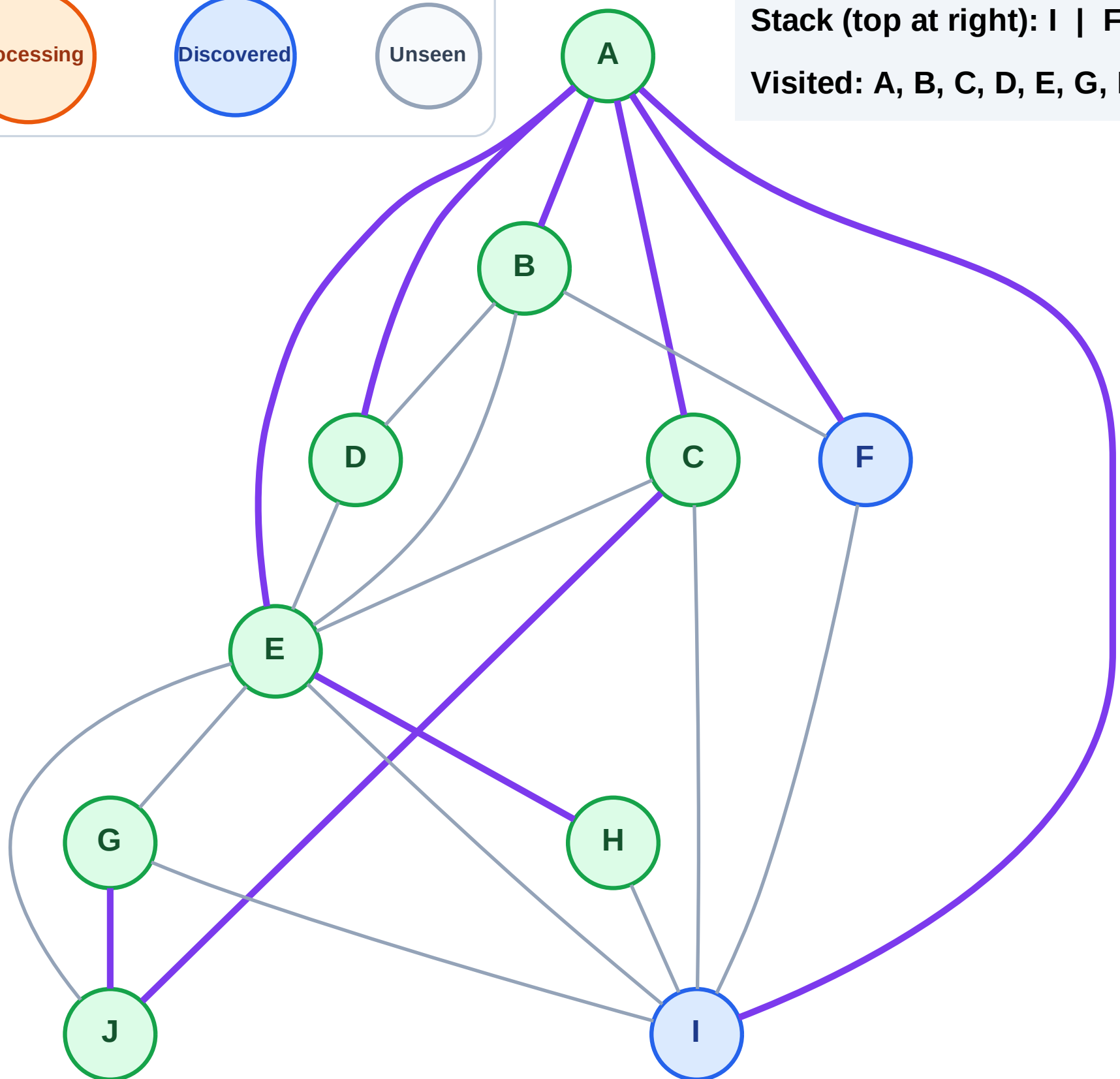
Step 17: No new neighbors from H

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, G, H, J



DFS Traversal

Step 18: Process node F

Legend

Complete

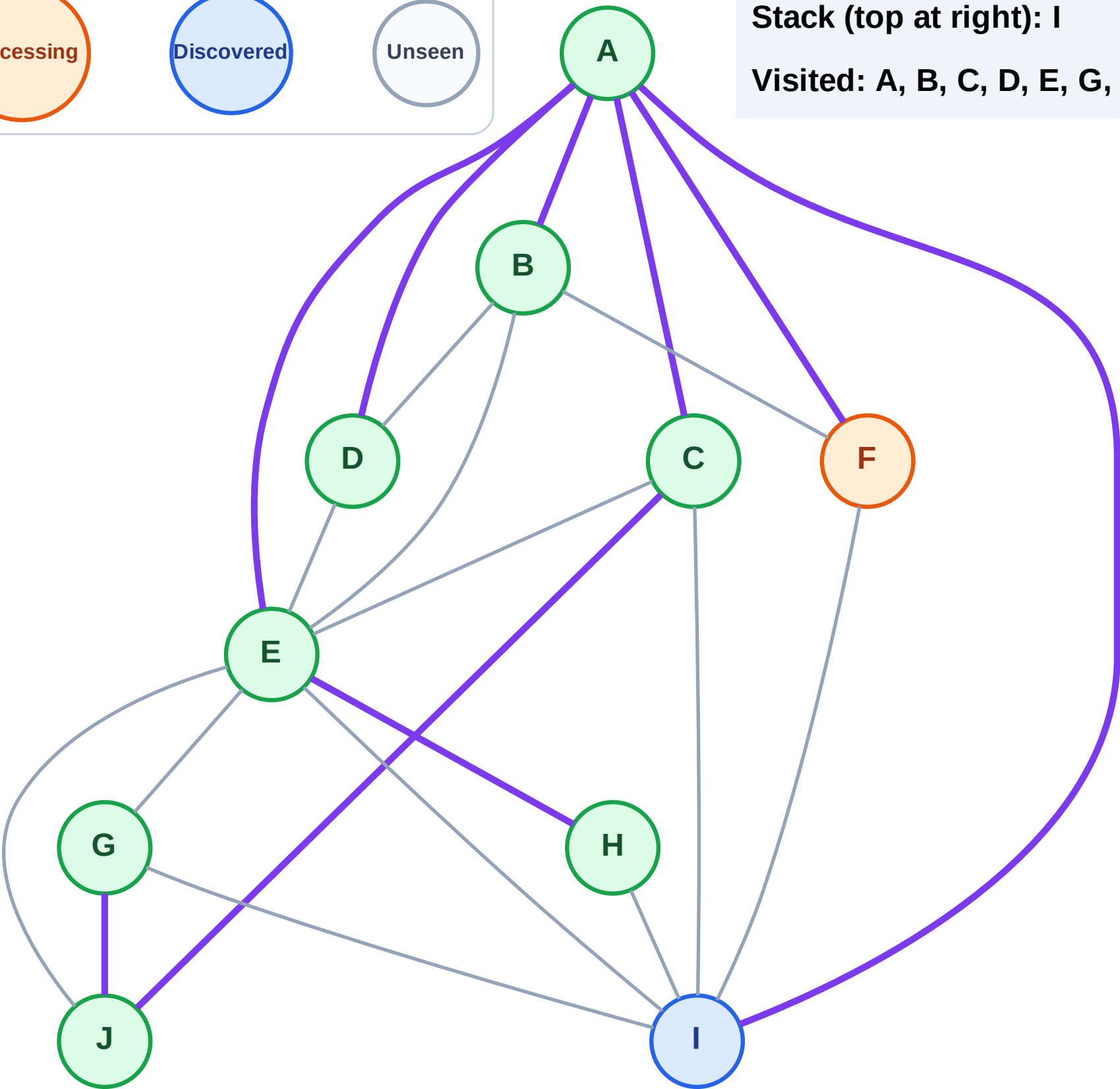
Processing

Discovered

Unseen

Stack (top at right): I

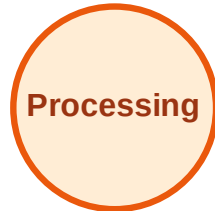
Visited: A, B, C, D, E, G, H, J



DFS Traversal

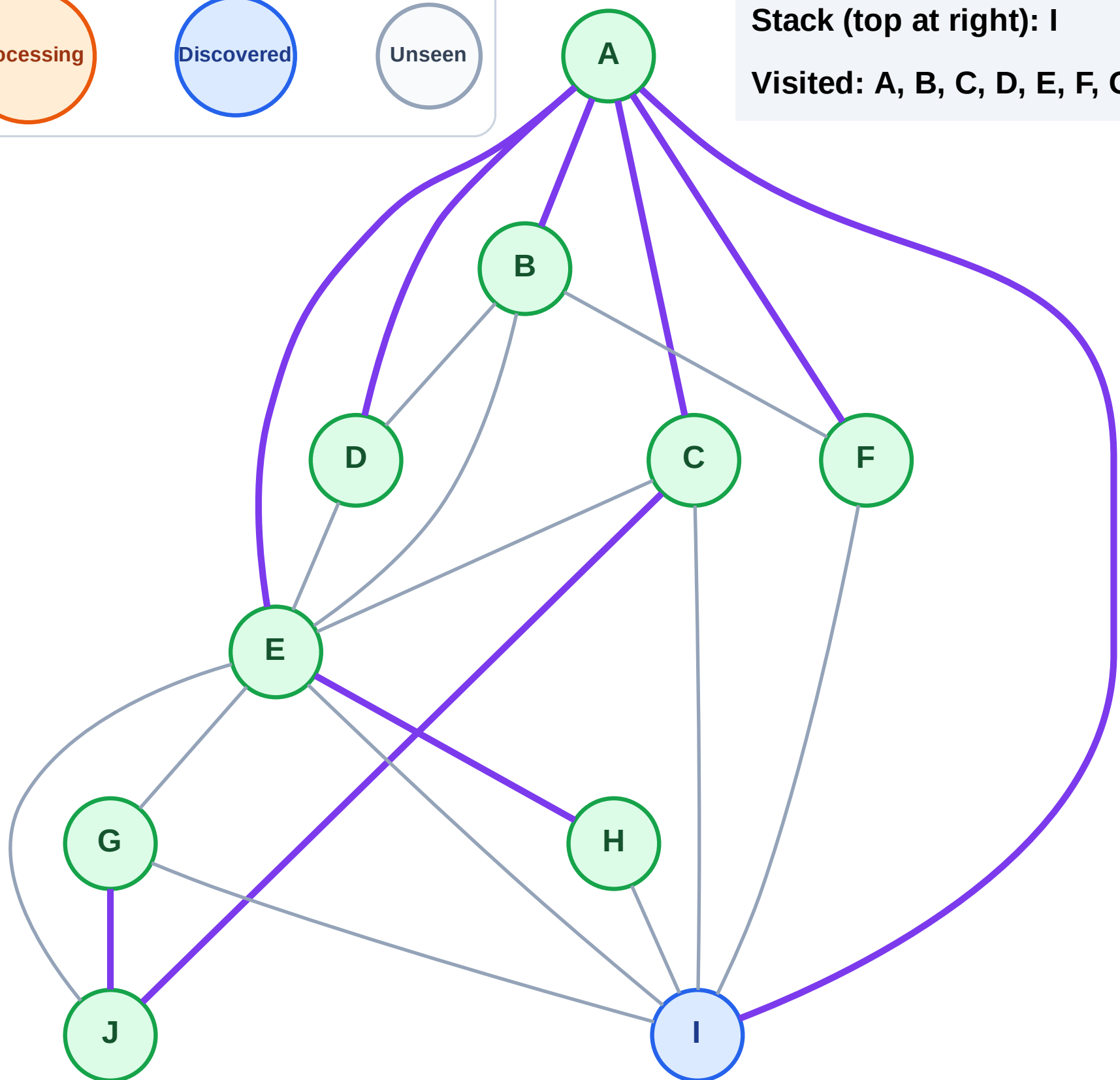
Step 19: No new neighbors from F

Legend



Stack (top at right): I

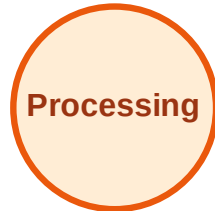
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

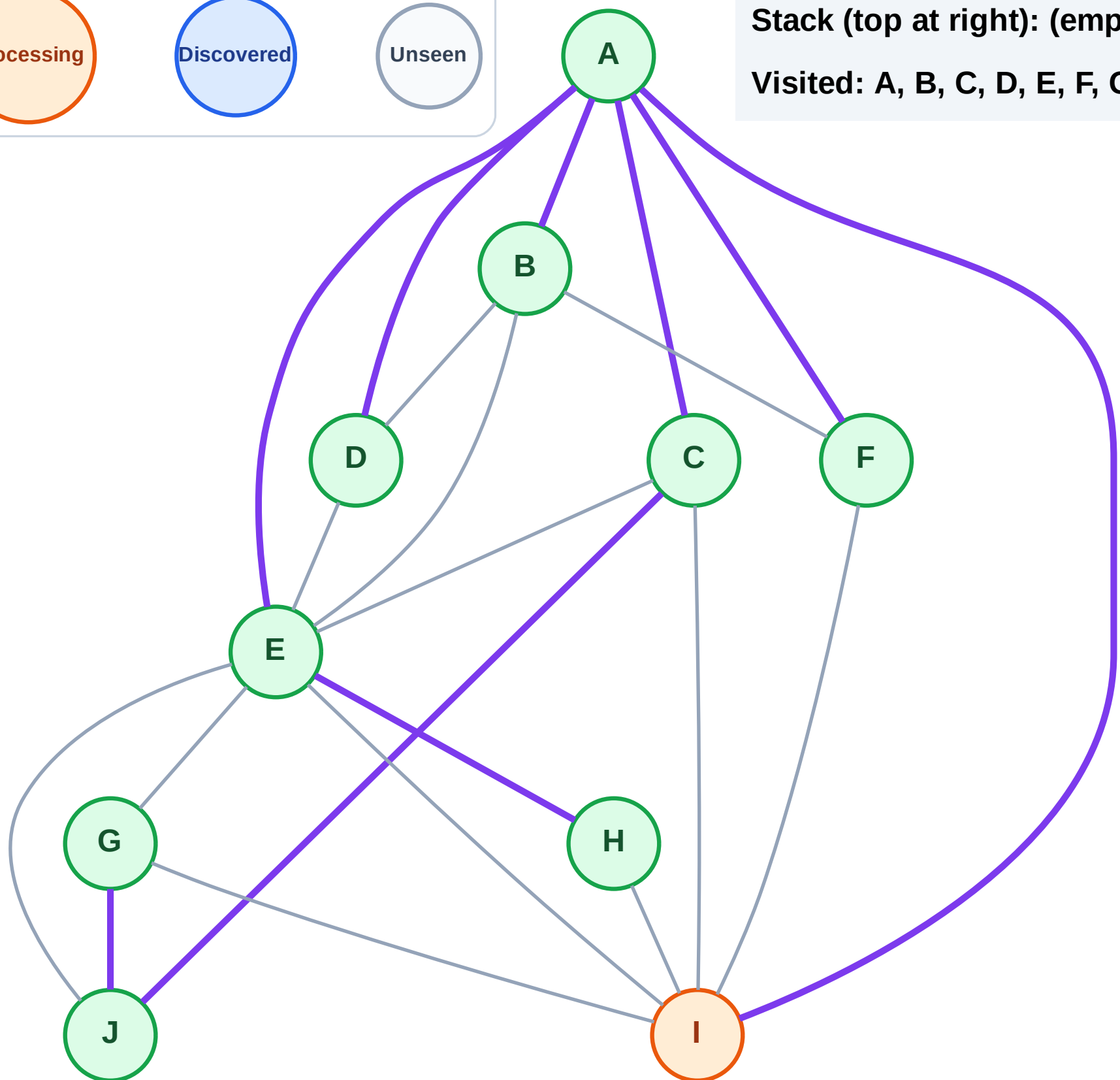
Step 20: Process node I

Legend



Stack (top at right): (empty)

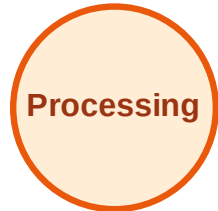
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

